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RESEARCH ARTICLE

Adaptive Buffering Strategies for Incremental Learning Under Concept Drift in Lifestyle Disease Modeling

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ABSTRACT Lifestyle diseases such as diabetes manifest through subtle and non-stationary clinical patterns, posing significant challenges for real-time prediction and monitoring. Conventional machine learning models often struggle to maintain performance under evolving data distributions due to concept drift. This study proposes an adaptive deep learning framework designed to handle concept drift through incremental learning in clinical data streams. The investigation centers on evaluating the effectiveness of various buffering strategies namely, adaptive buffering, FIFO buffering, and streaming without buffering combined with drift detection mechanisms for healthcare prediction. A balanced clinical dataset exhibiting evolving patterns was used to benchmark model performance. Deep learning architectures, including BiLSTM, GRU, LSTM, and Bayesian Neural Networks were incrementally trained, and their drift (Abrupt, Gradual and Recurring) responsiveness was assessed using three categories of detectors: 1) statistical test-based, 2) error-rate-based, and 3) uncertainty-based approaches leveraging Monte Carlo Dropout. Results indicated that adaptive buffering strategies consistently outperformed FIFO and no-buffer strategies, yielding higher accuracy, precision, and recall, especially under abrupt and large drift magnitudes. The hybrid drift detection method, combined with Bi-LSTM, demonstrated the best performance in maintaining retention and minimizing forgetting, even as the drift magnitude increased. Additionally, the drift magnitude study highlighted that larger drifts had a significant impact on model performance, with adaptive buffering and uncertainty-based drift detection proving to be more resilient to high drift intensities. This research underscores the importance of combining robust drift detection methods and adaptive buffering strategies to enhance the robustness of models dealing with concept drift in real-world applications.

INDEX TERMS Adaptive buffer, Bayesian neural network (BNN), bi-directional LSTM (BiLSTM), centers for disease control and prevention (CDC), concept drift detection and adaptation, first in first out (FIFO), gated recurrent unit (GRU), incremental learning, long-short term memory (LSTM), national health and nutrition examination survey (NHANES), uncertainty estimation.

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I. INTRODUCTION

Over the past decade, there has been a significant shift in the global health landscape, with communicable diseases

giving way to non-communicable diseases (NCDs) that are becoming increasingly prevalent as a result of lifestyle choices. According to the World Health Organization [1], non-communicable diseases are responsible for the deaths of around 41 million people each year, which contributes to 74% of all deaths that occur worldwide. When it comes to these, the worldwide mortality burden is dominated by malignancies, chronic respiratory illnesses (4.1 million), diabetes (1.5 million), and cardiovascular diseases (17.9 million). Several of these disorders have their origins in lifestyle variables that can be altered, such as unhealthy eating habits, a lack of physical activity, the use of tobacco products, the abuse of alcohol, chronic stress, and sleep cycles that are interrupted.

Emerging economies and cultures that are in the process of transitioning are experiencing an even more complicated burden. There are a number of factors that contribute to lifestyle disruptions that appear as chronic metabolic or vascular disorders [2]. These factors include migration from rural to urban areas, fast industrialization, changing sociocultural norms, and the combined hazards of under-nutrition and obesity. As a result of increased urbanization, even those that have historically been considered to be at a low risk are experiencing an increase in the incidence rates of diabetes and hypertension.

In addition, chronic diseases are not merely clinical issues but also significant economic disruptor within communities, contributing to rising dependency ratios, diminished productivity, and increased financial strain on both households and healthcare systems. Addressing these challenges necessitates a paradigm shift toward proactive, personalized public health approaches focused on prevention and early detection. This section contextualizes the broader problem of lifestyle diseases, examines the limitations of applying static machine learning models to evolving health data, introduces the concept of concept drift, underscores the role of adaptive buffering techniques, and positions the study's contribution within the wider pursuit of continuous and individualized health prediction—drawing from representative clinical data sources where applicable.

A. THE NATIONAL HEALTH AND NUTRITION EXAMINATION SURVEY (NHANES) A STANDARD FOR MONITORING THE HEALTH OF THE POPULATION

One of the most extensive datasets for lifestyle and health behavior analytics is provided by the NHANES, which was first conducted in the early 1960s and has undergone continuous development ever since. Structured interviews, clinical evaluations, and laboratory investigations are the methods that are utilized by the NHANES, which is carried out by the CDC of the United States of America. The scope of this study encompasses socio-demographic factors, anthropometric measurements, food intake, physical activity, mental health, medication use, and biomarkers of disease.

As a result of its longitudinal richness and methodological integrity, the NHANES is considered to be the gold standard for epidemiological and predictive modeling. It captures real-world patterns that span many years over a wide range of racial, ethnic, and socioeconomic strata, which is a key aspect of researching the temporal evolution of risk factors for chronic diseases [3].

It is suitable for feature selection, deep learning, and population-wide insights, and the NHANES offers researchers in machine learning a one-of-a-kind opportunity to model dynamic health hazards by utilizing thousands of characteristics. Because of its cross-cycle comparability, it also makes it easier to research idea drift and adaptive learning. In these types of studies, models that were trained on earlier cycles are validated on subsequent cycles to evaluate their generalizability.

B. THE ROLE OF ARTIFICIAL INTELLIGENCE IN THE TRANSITION FROM GENERALIZED RISK TO PERSONALIZED PREDICTION

Regression-based tools, such as the Framingham Risk Score for cardiovascular diseases and the ADA (American Diabetes Association) risk test for diabetes, have been utilized extensively in traditional public health models for the purpose of risk prediction. When it comes to offering individualized, real-time insights that take into consideration the complex interconnections between variables such as age, ethnicity, comorbidities, behavior, and environmental exposure, these models are excellent at capturing average trends; nevertheless, they fall short when it comes to providing personalized insights.

A new dimension has been introduced here by the application of AI and ML. The goal of machine learning algorithms is to discover non-linear relationships, manage high-dimensional data, and continuously improve with fresh inputs. Some examples of these algorithms are random forests, support vector machines, gradient boosting, and deep neural networks. Studies have demonstrated that machine learning-based models perform better than traditional statistical techniques when it comes to predicting illnesses such as diabetes, hypertension, stroke, and chronic kidney disease when they are trained on big datasets such as the NHANES [4], [5].

AI models have the potential to adjust and customize risk estimates for specific patients, which is the primary advantage of using these algorithms. Using such models, doctors and public health officials can effectively focus interventions because they are able to incorporate patient-specific features in real-time that provide precision health tools. However, this capability comes with a catch: the majority of models that are currently available are static. They make the premise that there is a stable relationship between the inputs and the outputs, an assumption that is becoming increasingly problematic in this day and age of dynamic health data.

C. CONCEPT DRIFT: A HIDDEN CHALLENGE IN CHRONIC DISEASE MODELLING

It is a problem that is inherent in health data due to shifting population habits, environmental conditions, medical advancements, or policy changes [6]. Concept drift arises when the statistical correlations between predictors and outcomes experience a change over time. It is possible, for instance, that a diabetes risk model that was established in 2010 will not function effectively in 2025 due to changes in the average body mass index (BMI), dietary patterns, physical activity, or even new diagnostic thresholds put out by clinical recommendations. The detailed explanation of concept drift is discussed in section III.

In longitudinal and streaming datasets like the NHANES, these alterations are not anomalies but typical behavior. The failure to take them into consideration results in the degradation of the model over time, a reduction in accuracy, and the possibility of misclassification, which is especially hazardous in clinical settings.

D. INCREMENTAL LEARNING: TOWARD CONTINUAL ADAPTATION

Incremental learning has developed as a fundamental method in the field of machine learning as a means of overcoming the limits of static models. In contrast to batch learning, which necessitates doing a complete retraining of the model with all of the data, incremental learning involves continuously updating the model with new data as it becomes available. In the field of healthcare, this is especially useful in situations where data is collected over time, such as in the case of electronic health records (EHRs), wearable devices, mobile applications, or public surveillance applications. To maintain performance and scalability, incremental models can integrate developing trends such as new disease variants, seasonal effects, or behavioral shifts [7]. Incremental models can also incorporate novel disease variations. When applied to chronic diseases, incremental learning enables the following:

- Real-time, explainable forecasts for decision support
- Scalability to new populations or geographical areas
- Reduced computational costs in comparison to comprehensive retraining
- Timely updates while patient profiles are evolving

Despite this, the effectiveness of incremental learning is contingent on successful data memory management, which involves figuring out how to keep relevant previous data while getting rid of noise and redundant information.

E. ADAPTIVE BUFFERING MEMORY MANAGEMENT FOR SMART LEARNING

Adaptive buffering is a system that saves a subset of historical data in order to guide model updates. This gets us to the concept of adjustable buffering. Imagine it as a dynamic memory bank that assists the model in remembering pertinent information and forgetting irrelevant information. The following are some examples of buffering strategies:

- No Buffer: This is the quickest but most forgetful strategy; it just updates the model with the most recent batch
- First-in, first-out (FIFO) Buffer: This buffer keeps a stationary window of recent data, which frees up some RAM
- Adaptive Buffer: which stores only the drifted anomalies for retraining

Indicators such as uncertainty, drift magnitude, or error rates are used to determine which data should be retained by the adaptive buffer.

Adaptive buffers make it possible to be flexible in response to changes in the healthcare industry, where some alterations are gradual (for example, dietary patterns) and others are abrupt (for example, pandemic effects). A sudden increase in fasting glucose is noticed across all patients; for example, the model can preserve relevant batches and update weights accordingly, thus preserving accuracy without the need for constant full retraining every time.

In NHANES-based modeling, since each cycle brings new sampling methods, variables, and social contexts, this is of greater significance than in other modeling approaches. The model can maintain its clinical validity and contextual relevance over time due to adaptive buffering.

F. A TESTBED FOR CONCEPT DRIFT

When it comes to the research of idea drift, there are very few datasets that are as suitable as NHANES. It offers a one-of-a-kind platform equipped with cycles that span decades and thousands of features, allowing users to,

- Simulate non-stationary situations by comparing older cycles to more recent cycles
- Evaluate the performance of artificial intelligence in population-wide disease surveillance
- Explore buffer-based model adaptation
- Benchmark drift detection techniques using real-world data

Furthermore, the NHANES incorporates a wide range of subgroups on the basis of age, gender, ethnicity, education, and income, which enables researchers to explore drift heterogeneity across populations. It is important to make sure that models do not perpetuate prejudices or inaccuracies among underrepresented populations, which is in line with the increased emphasis on health equity [8].

G. RESEARCH OBJECTIVES AND STUDY CONTRIBUTION

In light of this, the following are the objectives of the presented research:

- 1) Create and test incremental learning algorithms that are capable of adjusting to non-stationary clinical data distributions
- 2) To develop and evaluate memory-efficient buffering solutions (No Buffer, FIFO, and Adaptive Buffer) that strike a balance between learning plasticity and maintaining stability

- 3) To demonstrate the efficacy of adaptive model updating in enhancing the early diagnosis of lifestyle-related diseases and the continuous monitoring such diseases
- 4) To provide a framework that is scalable for producing real-time, drift-aware personalized health modeling that has clinical and public health applications.

Not only does this study contribute to the creation of algorithms, but it also contributes to the use of translational artificial intelligence in the healthcare industry. It provides useful workflows, evaluation metrics, and insights for incorporating adaptive learning into surveillance systems.

The mind map of the presented work, which will give a graphical overview, is shown in Figure 1. To guide the reader through the presented study, the remainder of the manuscript is structured as follows. Section II reviews key literature on concept drift, incremental learning, and buffering mechanisms in healthcare and related domains. Section III and section IV introduce the mathematical formulations underpinning concept drift and incremental model adaptation, providing the theoretical foundation for understanding the presented research. In Section V, we detail three buffering strategies that are No Buffer, FIFO Buffer, and Adaptive Buffer, and analyze their implications for clinical learning. Section VI presents the proposed framework for drift-aware lifestyle disease management, integrating incremental learning with uncertainty-driven buffering. Section VII describes the dataset; target variables and model configurations used in this study are summarized in Section VIII. Section IX outlines the experimental setup, including drift simulation techniques and evaluation protocols. Section X reports and discusses the comparative performance of the buffering strategies. In Section XI, we provide clinical interpretations and case-level insights from our results. Finally, Section XII concludes the paper with a summary of key findings and outlines future research directions.

II. BACKGROUND AND RELATED WORK

The growing adoption of AI-based predictive models in healthcare brings forth the significant challenge of concept drift, where model performance degrades as underlying data distributions shift over time [9]. This phenomenon is a well-documented issue in streaming data applications, and a substantial body of research has been dedicated to methods for its detection and adaptation [10]. While innovative approaches for drift adaptation, such as trust decay-based temporal learning, have been explored in other domains like recommender systems [11], the healthcare field presents unique difficulties. A particularly critical problem is the prediction of performance drift in clinical settings where ground truth labels are often delayed or entirely unavailable, making it essential to develop methods that can ensure model reliability without immediate outcome data [12].

Chronic illnesses, which include cardiovascular disease (CVD), diabetes, and hypertension, continue to be among the leading causes of morbidity and mortality across the globe. Machine learning (ML) and artificial intelligence (AI)

have been incorporated into the healthcare industry, which has resulted in the opening of new doors for the prediction and management of these disorders. Providing complete health and nutritional data that is typical of the population of the United States. The purpose of this literature review is to investigate recent developments in machine learning and artificial intelligence applications for predicting chronic disease risk. The review focuses on works that have been published in Scopus Q1 journals since 2024.

A. MACHINE LEARNING APPLICATIONS IN CHRONIC DISEASE PREDICTION

1) HYPERTENSION PREDICTION

To predict hypertension, [13] built a model based on machine learning by assessing environmental chemical exposures using data from 2003 to 2016. In order to handle missing data, the study applied multiple imputation approaches. Additionally, this study utilized a variety of machine learning algorithms in order to evaluate the impact of environmental chemicals on the risk of hypertension. The findings proved the potential of machine learning models in identifying individuals who are at a high risk of developing hypertension and brought to light the role of environmental factors in the development of hypertension.

2) CARDIOVASCULAR AND ALL-CAUSE MORTALITY

In a prospective study that was published in BMC Public Health [14], researchers evaluated the efficacy of machine learning models in predicting death from cardiovascular and all-cause causes by using data from 2007 to 2010. During the research, various techniques were utilized, including support vector machines (SVM), random forest, and extreme gradient boosting. The results showed that machine learning models performed better than traditional statistical methods, with AUC values of 0.862 and 0.836 for cardiovascular mortality and all-cause mortality, respectively. The research highlighted the significance of lifestyle choices in relation to mortality risk, as well as the potential of machine learning in relation to risk stratification.

3) METABOLIC SYNDROME PREDICTION

To develop and validate machine learning-based models for the prediction of metabolic syndrome (MetS), international cohort validation research was conducted. The research utilized a number of machine learning methods, such as multilayer perceptron (MLP) and XGBoost, to analyze the data obtained from the China Health and Retirement Longitudinal Study (CHARLS), the Korea National Health and Nutrition Examination Survey (KNHANES), the UK Biobank, and the NHANES. A high level of predictive ability was exhibited by the MLP model across a variety of cohorts, with an AUC of 0.9055 in the NHANES cohort. This work demonstrates that machine learning models can be applied to a wide variety of populations for the purpose of MetS prediction.

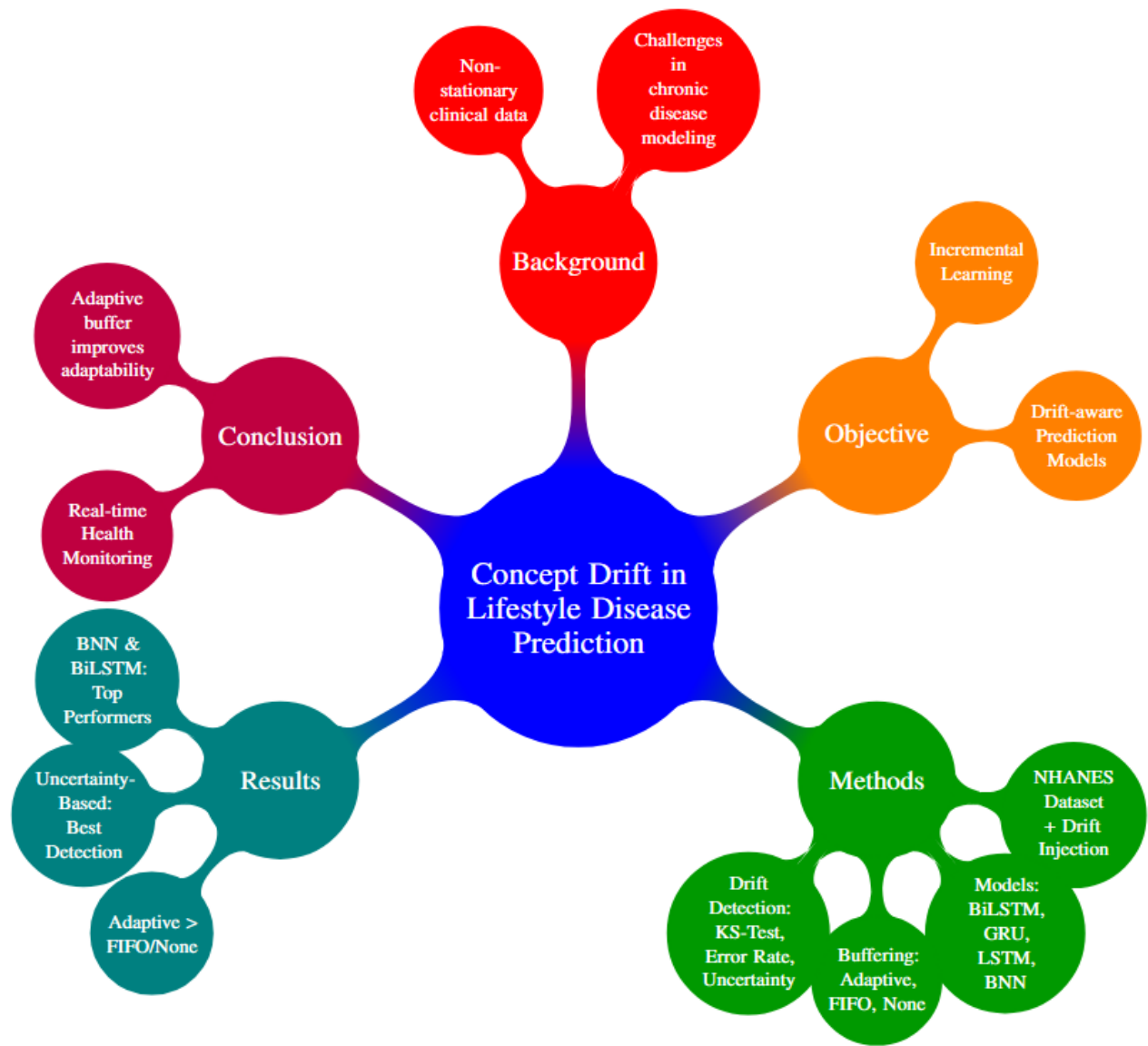


FIGURE 1. Mind map of the experimental framework for concept drift detection in lifestyle disease prediction.

4) ATHEROSCLEROTIC CARDIOVASCULAR DISEASE (ASCVD) IDENTIFICATION

The primary objective of [15] was to construct an interpretable machine learning model for the diagnosis of ASCVD by utilizing demographic factors and dietary patterns derived from data spanning the years 1999 to 2018. In addition to demonstrating that the incorporation of dietary habits in addition to demographic parameters improved the predictive accuracy of ASCVD risk models, the study highlighted the significance of model interpretability in clinical settings.

5) DIABETES COMPLICATIONS PREDICTION

Researchers in Aotearoa New Zealand used machine learning (ML) and social administrative data to make predictions

about the likelihood of problems associated with diabetes. The purpose of the study was to address ethnic disparities in health outcomes related to diabetes. The research highlighted the potential of merging machine learning with administrative data in order to identify individuals who are at high risk and to use this information to inform targeted interventions, particularly in communities that are facing health disparities [16].

6) OSTEOPOROSIS RISK ASSESSMENT

An ML model for diagnosing osteoporosis was built by [17] that was published in BMC Research Notes. The model was developed using data from the data set. The performance of the model was superior to that of conventional clinical assessment tools, and it highlighted the significance of

interpretable AI techniques. In addition, the study emphasized the necessity of integrating various data sources, such as genetic information and imaging investigations, in order to improve the robustness of predictive models.

7) CHRONIC KIDNEY DISEASE (CKD) IN ABDOMINAL OBESITY

Using machine learning techniques, [18] researched the identification and optimization of key factors for chronic kidney disease (CKD) in individuals who had abdominal obesity. Through the utilization of data spanning from 2005 to 2018, the research endeavor employed a number of machine learning techniques to discover critical predictors. These predictors were triglyceride glucose, waist circumference, and the composite dietary antioxidant index. Machine Learning was shown to be helpful in revealing the intricate relationships between metabolic variables and the risk of chronic kidney disease (CKD) by the findings.

8) ISCHEMIC STROKE AND HEAVY METAL EXPOSURE

Using data from 2003 to 2018, [19] utilized machine learning techniques to investigate the connection between heavy metal exposure and ischemic stroke. The promise of machine learning in environmental health research was demonstrated by the study, which focused on identifying connections between heavy metal levels and stroke risk. The study utilized logistic regression and other machine learning techniques.

9) CONCEPT DRIFT AND INCREMENTAL LEARNING IN HEALTHCARE

To learn changing data, [20] presented a neighbor-searching discrepancy-based drift detection system. Real concept drift, which refers to changes in the categorization boundary over time, was the subject of the study, which addressed the difficulties associated with recognizing it. The proposed method displayed a high level of accuracy in detecting real concept drift while disregarding virtual drift. This provided useful insights for sustaining the performance of machine learning models in healthcare environments that are constantly changing. CEL is a continual learning model for disease outbreak prediction that was presented by [21]. This model makes use of domain adaptability through elastic weight consolidation. The purpose of the model was to reduce the risk of catastrophic forgetting in deep neural networks by punishing changes to parameters that were considered to be relevant. During the evaluation of CEL's performance, diseases such as measles, mpox, and influenza were used to demonstrate its adaptability to incremental data and its potential for proactive disease control.

10) DOMAIN-INCREMENTAL ADAPTATION

PAGE (Patient Activation in Genomics and Engagement) is a domain-incremental adaptation approach with past-agnostic generative replay that was introduced by [22] to offer intelligent healthcare. To strike a balance between domain

adaptation and knowledge retention, PAGE makes it possible to do generative replay without preserving data from previous domains. An expanded inductive conformal prediction method is incorporated into the model in order to provide confidence scores and credibility values, which in turn improves the interpretability of disease detection.

11) BIAS AND EQUITY IN MACHINE LEARNING APPLICATIONS

In their scoping study, [14] investigated the presence of bias in machine learning systems that were designed to address non-communicable diseases at the population level. To ensure that healthcare results are equal, the study emphasized the significance of considering biases during the data collection, model building, and deployment processes. Transparency and fairness were emphasized as being equally important.

12) ADVANCED MACHINE LEARNING TECHNIQUES IN CHRONIC DISEASE PREDICTION

With the help of data, [23] investigated the use of a variety of supervised machine learning algorithms to make predictions about depressive illnesses. In this study, a comparison was made between several different statistical methods, including logistic regression, random forest, Naïve Bayes, support vector machines (SVM), eXtreme Gradient Boost (XGBoost), and Light Gradient Boosted Machine (Light-GBM2). Shapley Additive Explanations (SHAP) were utilized to explain the significance of the features in the models that were examined. The findings indicated that machine learning models were capable of accurately predicting depressive disorders, with certain algorithms exceeding standard statistical methods in terms of accuracy and interpretability.

Using data from 2007 to 2016, [24] built a predictive model for teenage metabolic syndrome (MetS). The researchers used LASSO (Least Absolute Shrinkage and Selection Operator) regression to choose features and constructed numerous machine learning models, one of which was LightGBM, which produced the greatest performance with an area under the curve (AUC) of 0.969. Both of these models were used to pick features. In addition, the model was interpreted further by employing SHAP values, which led to the identification of critical predictors such as the age-specific proportion of body mass index (BMI), weight, and upper arm circumference. In the study, the potential of machine learning models in early, non-invasive screening for metabolic syndrome in adolescents was highlighted.

13) LIFESTYLE FACTORS AND OBESITY

Utilizing pooled data from the China Health and Nutrition Survey (CHNS) and the National Health and Nutrition Examination Survey, research conducted by Guo et al. utilized interpretable machine learning techniques to evaluate the relative significance of lifestyle factors in predicting overweight and obesity among people. The study utilized decision tree, random forest, and gradient-boosting decision

tree models. The SHAP analysis revealed that sedentary behavior, alcohol consumption, and protein intake were important predictors of overweight and obesity. The study was conducted in the United States. Based on these findings, it appears that machine learning models have the potential to offer detailed insights into lifestyle-related health concerns.

B. CONCEPT DRIFT DETECTION AND INCREMENTAL LEARNING IN HEALTHCARE

An innovative approach to identifying true idea drift in changing data streams was presented by [13], with a particular emphasis on changes in categorization boundary regions. The neighbor-searching discrepancy-based drift detection scheme that has been suggested is able to differentiate between real and virtual drift in an efficient manner, hence providing insights into the direction in which classification boundary alterations are occurring. Through the accurate identification of when model changes are required, this strategy improves the adaptability of machine learning models in healthcare environments that are constantly changing.

Reference [24] presented an overview of performance-aware drift detectors and emphasized the significance of monitoring model performance degradation in order to identify concept drift. In the study, several drift detection methods were grouped according to their mathematical definitions and tactics. This resulted in the creation of a consolidated taxonomy that can be used to comprehend and manage concept drift in predictive systems. It is possible for data distributions in healthcare applications to vary over time, and this framework helps to preserve the trustworthiness of machine learning models in those applications.

C. SUMMARY OF LITERATURE REVIEW AND FUTURE DIRECTIONS

Incorporating machine learning and artificial intelligence approaches into the analysis of data has resulted in major advancements in both the prediction and understanding of chronic diseases. Recent research has shown that various machine learning models are effective in predicting conditions such as hypertension, metabolic syndrome, depression, and obesity. These models frequently beat classic statistical methods in this regard. Furthermore, the creation of interpretable models and the use of SHAP values have contributed to an increase in the clinical applicability and transparency of these predictive tools.

Concurrently, developments in concept drift detection and incremental learning are addressing the issues that are posed by the ever-changing data distributions in the healthcare industry. When it comes to preserving the accuracy and dependability of machine learning models over time, it is essential to make use of innovative methods such as neighbor-searching discrepancy-based drift detection and performance-aware drift detectors.

The integration of these advanced machine learning approaches with real-time data streams, the enhancement of model adaptability, and the guaranteeing of fair healthcare

results across a wide range of people should be the primary interests of future study. Additionally, the development of standardized frameworks for model evaluation and drift detection will be vital for the continued success of machine learning applications in chronic illness risk modeling. This is because these frameworks will prevent drift.

D. SCOPE AND LIMITATION OF THE WORK

This study explores the amalgamation of various temporal and spatial deep learning models with adaptive buffering mechanisms to detect and respond to concept drift in the context of lifestyle disease modeling. Using the dataset, it systematically benchmarks three drift detection strategies, such as error-based, statistical test-based, and uncertainty-based, across multiple deep learning models, namely Bi-LSTM, GRU, LSTM, and Bayesian Neural Networks (BNNs). The scope can be extended to,

- Assessing how streaming batch-based model updates can be effectively supported with different buffer strategies to sustain predictive performance over time
- Examining the role of adaptive buffering in retaining clinically relevant samples and improving responsiveness to non-stationary health data
- Comparative understanding of how different drift detectors perform under non-stationary data, both in terms of sensitivity and clinical interpretability
- Simulating synthetic concept drift within real-world health data streams to mirror clinically unexpected behavior changes in patient populations
- Emphasizing uncertainty-aware modeling with MC Dropout and BNNs for more explainable and stable performance in dynamic healthcare environments

The limitations of the presented research are,

- The study uses synthetically induced drift scenarios to evaluate detection mechanisms. Though the drift was induced in the controlled manner, still the full essence of complete non-stationarity may not have been captured
- Clinically significant modalities such as imaging, wearable sensors, or genomics that could influence drift dynamics are not modeled in this study, as the focus is on the tabular nature of dataset
- Live electronic health record (EHR) systems are not used in the experimentation, but the dataset is still taken as stream batches to emulate the streaming environment, which might limit the full scope of Online learning
- The study is based on U.S.-specific health survey data, so adapting to the different demography requires changes to the preprocessing phase of the learning models
- The use of complex architectures like BNNs with MC Dropout introduces computational overhead that may limit the deployment in low-resource environments

III. CONCEPT DRIFT

Lifestyle diseases such as diabetes, hypertension, and cardiovascular disorders are not static conditions; they are shaped

by continuously evolving behavioral, environmental, and socioeconomic factors. These dynamic influences introduce *non-stationarity* in the learning environment, known as **concept drift**, into clinical and lifestyle datasets, where the underlying data distributions change over time. Such drift poses a significant challenge to traditional predictive models, which operate under the assumption of static distributions and are therefore prone to degradation in accuracy as real-world conditions evolve. In streaming healthcare environments, this drift manifests through changes in patient habits, healthcare access, seasonal effects, and public health interventions. To address this, **incremental learning** provides a promising solution by enabling models to update continuously as new data becomes available. However, the efficacy of this approach heavily depends on how historical data is managed. This motivates the incorporation of **buffering strategies** such as FIFO (First In First Out) or adaptive memory modules that selectively retain relevant information to support model adaptability and mitigate the effects of concept drift in personalized lifestyle disease risk prediction. The mathematical perspective of the concept drift is discussed next.

Let $X \in \mathcal{X}$ be the input space and $Y \in \mathcal{Y}$ be the output space. In supervised learning, the data is assumed to follow a joint distribution shown in equation 1 as,

$$\mathcal{D}_t = P_t(X, Y) \quad (1)$$

Concept drift occurs when the joint distribution changes over time as shown in equation 2 as,

$$\exists t_1, t_2 \text{ such that } \mathcal{D}_{t_1} \neq \mathcal{D}_{t_2} \quad (2)$$

TABLE 1. Mathematical types of concept drift.

Type	Description	Mathematical Expression
Covariate Drift	Change in input distribution	$P_{t_1}(X) \neq P_{t_2}(X)$
Prior Drift	Change in class priors	$P_{t_1}(Y) \neq P_{t_2}(Y)$
Real Drift	Change in decision boundaries	$P_{t_1}(Y X) \neq P_{t_2}(Y X)$

There are various types of drift available based on what is changing in the data distribution summarized in table 1. The drift can be,

- 1) Changes in the marginal distribution $P(X)$ called as **covariate drift**.
- 2) Changes in the posterior distribution $P(Y|X)$ called as **real drift**.
- 3) Changes in class priors $P(Y)$ called as **prior probability drift**.

The presented work aims at capturing real drift in the dataset using incremental learning with buffering strategies.

IV. INCREMENTAL LEARNING FRAMEWORK

The rising prevalence of lifestyle-related diseases such as obesity, type 2 diabetes, and cardiovascular conditions highlights the quintessential need for predictive healthcare

systems that are both responsive and adaptive. These diseases are closely related to factors that fluctuate over time—dietary habits, physical activity levels, stress, and environmental influences—making the associated health data inherently dynamic. As patient behaviors shift, so too do the patterns within clinical datasets, creating a moving target for traditional predictive models. Capturing this temporal variability is essential for accurate risk identification and timely intervention. To address this challenge, the proposed framework employs an incremental learning approach designed to adapt to these evolving data patterns. Figure 2 illustrates the proposed architecture for incremental learning under concept drift using buffered clinical data streams. Streaming data, such as patient records arriving over time, are first passed through a buffer mechanism that temporarily stores incoming data in batches. These buffered batches enable both short-term memory and statistical analysis. The buffered data then undergo drift detection, which assesses whether significant changes have occurred in the data distribution. When drift is detected, the model is updated incrementally using the buffered data, thereby ensuring adaptability to evolving clinical patterns. In parallel, the current buffered instance is used to generate real-time predictions for clinical decision support. A feedback loop ensures that the model remains synchronized with observed data shifts, supporting continuous learning in dynamic healthcare environments.

Let $\mathcal{D}_t = \{(x_t, y_t)\}$ the data be at time t . In incremental learning, the model f_t is updated over time as shown in equation 3.

$$f_{t+1} = \mathcal{A}(f_t, (x_t, y_t)) \quad (3)$$

where $\mathcal{A}$ denotes the learning algorithm capable of adapting the model incrementally.

V. BUFFERING STRATEGIES FOR DRIFT-AWARE CLINICAL INCREMENTAL LEARNING

In streaming healthcare environments, where patient data arrives sequentially and may undergo distributional changes over time. The ability of a model to retain or discard past information is critical for maintaining predictive reliability. Buffering strategies play a central role in managing this temporal information by determining which data points are preserved for future learning updates. In this section, we explore three such strategies: No Buffer, FIFO Buffer, and Adaptive Buffer, each offering a distinct approach to balancing memory efficiency, responsiveness to concept drift, and clinical relevance. These strategies are particularly important in the context of lifestyle disease modeling, where patient behaviors and risk factors evolve gradually or abruptly over time. We examine each of these buffer strategies in detail below.

A. NO BUFFER STRATEGY

The *no-buffer* strategy is the most memory-efficient incremental learning setup, where the model is updated only on the most recent batch of data. This approach is suitable

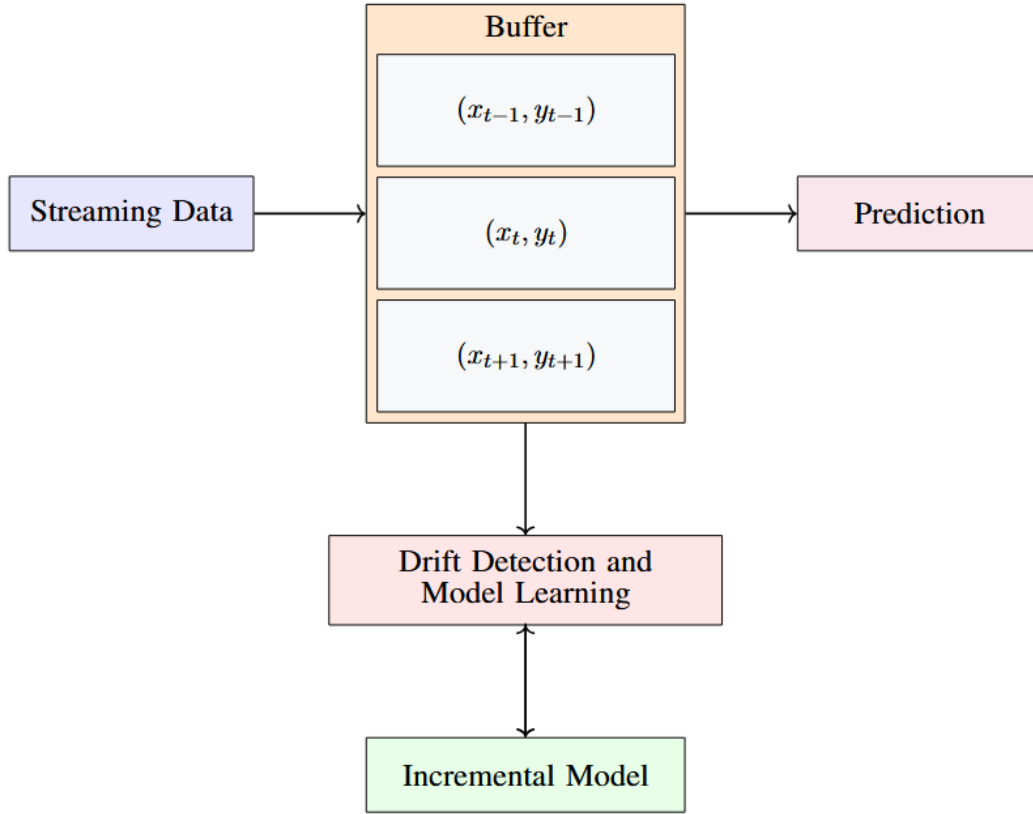


FIGURE 2. General architecture of incremental learning with buffer-based drift adaptation.

for latency-sensitive or resource-constrained applications. However, it lacks historical context, making it prone to **catastrophic forgetting** and overfitting to short-term noise.

Let $\mathcal{B}_t$ denote the incoming batch at time t . The model f_θ is updated as given by equation 4

$$\theta_{t+1} \leftarrow \theta_t - \eta \cdot \nabla_{\theta} \mathcal{L}(f_{\theta}(\mathcal{B}_t), y_t) \quad (4)$$

where η is the learning rate and $\mathcal{L}$ is the loss function. This formulation can lead to unstable updates due to the small batch size and lack of memory.

The *No Buffer* strategy updates the model parameters using only the current incoming batch $\mathcal{B}_t \in \mathbb{R}^{n \times d}$, where n is the batch size and d is the feature dimension. The model update is given by equation 5,

$$\theta_{t+1} \leftarrow \theta_t - \eta \cdot \nabla_{\theta} \mathcal{L}(f_{\theta}(\mathcal{B}_t), y_t) \quad (5)$$

The overall space complexity induced by this approach is given by equation 6,

$$\mathcal{O}(\mathcal{M}_t) = \mathcal{O}(n) \quad (6)$$

The overall computational complexity induced by this approach is given by equation 7,

$$\mathcal{O}(C_t) = \mathcal{O}(n \cdot d \cdot h) \quad (7)$$

where h is the number of trainable parameters. This method is memory-light and fast but lacks resilience to long-term or recurring drifts due to the absence of historical data.

B. FIFO BUFFER STRATEGY

The *First-In-First-Out (FIFO)* buffer maintains a fixed-size window of the most recent data batches. When new data arrives and the buffer is full, the oldest batch is discarded. This buffer allows short-term memory, reducing variance and improving generalization.

For a buffer of size K , the active memory at time t given by equation 8,

$$\mathcal{M}_t = \{\mathcal{B}_{t-K+1}, \dots, \mathcal{B}_t\} \quad (8)$$

Model update is then carried out over this buffer given by 9,

$$\theta_{t+1} \leftarrow \theta_t - \eta \cdot \nabla_{\theta} \mathcal{L}(f_{\theta}(\mathcal{M}_t), y_t) \quad (9)$$

The *FIFO Buffer* strategy maintains a sliding window of the most recent K batches given by equation 10

$$\mathcal{M}_t = \{\mathcal{B}_{t-K+1}, \dots, \mathcal{B}_t\} \quad (10)$$

The model is retrained on this memory buffer given by equation 11,

$$\theta_{t+1} \leftarrow \theta_t - \eta \cdot \nabla_{\theta} \mathcal{L}(f_{\theta}(\mathcal{M}_t), y_t) \quad (11)$$

The overall space complexity induced by FIFO buffer approach is given by the equation 12,

$$\mathcal{O}(M_t) = \mathcal{O}(K \cdot n) \quad (12)$$

The overall computational complexity induced by the FIFO buffer approach is given by equation 13,

$$\mathcal{O}(C_t) = \mathcal{O}(K \cdot n \cdot d \cdot h) \quad (13)$$

This buffer allows short-term memory and smoothing but does not distinguish between informative and redundant samples.

C. ADAPTIVE BUFFER STRATEGY

The *adaptive buffer* dynamically adjusts its memory based on drift severity or model uncertainty. Unlike FIFO, it selectively retains data from batches where drift is detected, allowing focused retraining while minimizing redundancy. In adaptive replay buffer strategies, the model dynamically retains only those data batches that correspond to significant concept drift events. Unlike fixed-size buffers that store data indiscriminately, the adaptive buffer uses a drift detection mechanism typically based on uncertainty estimates such as Monte Carlo dropout to determine whether the current batch reflects a distributional shift. When a drift is detected, the corresponding batch is stored in memory and used for retraining. Batches that do not show any significant deviation from the previous distribution are discarded to conserve memory. This ensures that the model selectively learns from informative instances while avoiding overfitting to redundant or stationary data. The following figure 3 illustrates this process.

Let's $\mathcal{U}_t$ denote model uncertainty at time t . A drift is detected if it satisfies the equation 14

$$\mathcal{U}_t > \alpha \cdot \mathcal{U}_{t-1} \quad (14)$$

where $\alpha > 1$ is a sensitivity threshold. If drift is detected, the batch $\mathcal{B}_t$ is added to the buffer $\mathcal{A}_t$, and the model is updated using equation 15

$$\theta_{t+1} \leftarrow \theta_t - \eta \cdot \nabla_{\theta} \mathcal{L}(f_{\theta}(\mathcal{A}_t), y_t) \quad (15)$$

The *Adaptive Buffer* stores only those batches identified as conceptually novel based on predictive uncertainty. A drift is declared if equation 16,

$$\mathcal{U}_t > \alpha \cdot \mathcal{U}_{t-1}, \quad \alpha > 1 \quad (16)$$

Here $\mathcal{U}_t$ is the mean uncertainty score at time t . Let's $\mathcal{A}_t \subseteq \bigcup_{i=1}^t \mathcal{B}_i$ denote the set of batches corresponding to detected drifts. The model is updated using equation 17

$$\theta_{t+1} \leftarrow \theta_t - \eta \cdot \nabla_{\theta} \mathcal{L}(f_{\theta}(\mathcal{A}_t), y_t) \quad (17)$$

The overall space complexity induced by the FIFO buffer approach is given by the equation 18,

$$\mathcal{O}(M_t) = \mathcal{O}(m \cdot n), \quad m \leq t \quad (18)$$

The overall computational complexity induced by the FIFO buffer approach is given by equation 19,

$$\mathcal{O}(C_t) = \mathcal{O}(m \cdot n \cdot d \cdot h) \quad (19)$$

This strategy allows for dynamic, relevance-driven memory allocation, making it particularly effective for handling abrupt and recurring drifts. The aforementioned strategies' computational details are summarized in table 2.

Theorem 1: [Convergence Condition] Let $\{D_t\}_{t=1}^{\infty}$ denote a sequence representing the magnitude of concept drift at time t , and let the Adaptive Replay Buffer have a maximum capacity of B_{adaptive} . Suppose drift detection and retraining are triggered whenever $D_t > \tau$, for a fixed threshold $\tau > 0$. If the expected drift magnitude is bounded such that

$$\lim_{t \rightarrow \infty} \mathbb{E}[D_t] < \tau, \quad (20)$$

then the adaptive buffer stabilizes, and the system converges to a steady state with bounded memory growth.

Proof: The adaptive replay buffer adds data to memory only when a drift is detected, i.e., when $D_t > \tau$. Over time, if the expected drift magnitude satisfies the bound in Equation (20), the frequency of drift detection decreases.

By Markov's inequality given by equation 21,

$$\mathbb{P}(D_t > \tau) \leq \frac{\mathbb{E}[D_t]}{\tau} \quad (21)$$

implying that the probability of drift detection becomes arbitrarily small as $\mathbb{E}[D_t] \rightarrow 0$. As a result, fewer new samples are added to the buffer, and the update frequency diminishes.

This behavior leads to eventual stabilization of the buffer size and model parameters, assuming the underlying learning dynamics also hold. Hence, the adaptive system converges in both memory and predictive stability. $\square$

VI. PROPOSED FRAMEWORK FOR LIFESTYLE DISEASE MANAGEMENT

Figure 4 presents the proposed experimental workflow for evaluating adaptive buffering strategies under concept drift using the dataset. The process begins with a data preprocessing pipeline that performs data cleaning, exploratory analysis, and class balancing to ensure model-ready input. To simulate real-world non-stationarity, a drift induction pipeline introduces abrupt changes by manipulating class labels, effectively mimicking shifts in patient behavior or diagnostic criteria. The data is then streamed in mini-batches to emulate an online learning environment, where various deep learning architectures, including LSTM, GRU, Bi-LSTM, and Bayesian Neural Networks, are trained using one of three buffering strategies: No Buffer, FIFO Buffer, or Adaptive Buffer. A drift detection and adaptation pipeline monitors model behavior using statistical tests, hybrid approach and uncertainty measures to detect drift points, after which the model is selectively retrained on relevant batches, which is highlighted in the dotted red-lined path. Throughout the process, performance parameters such as

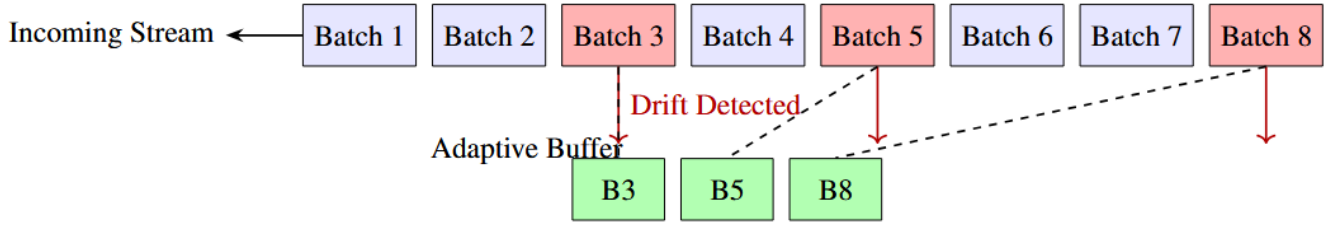


FIGURE 3. Illustration of the adaptive replay buffer— Only batches associated with significant drift (highlighted in red) are retained in memory for retraining. Other non-drift batches are discarded, allowing the buffer to adapt dynamically to changes in data distribution.

TABLE 2. Theoretical complexity comparison of buffer strategies.

Strategy	Space Complexity $\mathcal{O}(M_t)$	Compute $\mathcal{O}(C_t)$	Buffer Growth
No Buffer	$\mathcal{O}(n)$	$\mathcal{O}(n \cdot d \cdot h)$	Constant
FIFO Buffer	$\mathcal{O}(K \cdot n)$	$\mathcal{O}(K \cdot n \cdot d \cdot h)$	Fixed size
Adaptive Buffer	$\mathcal{O}(m \cdot n)$	$\mathcal{O}(m \cdot n \cdot d \cdot h)$	Drift-triggered

accuracy values and uncertainty values are measured to assess each buffering strategy's effectiveness. The final stage involves recording key performance metrics such as ROC, AUC, F1-score, accuracy, and loss, allowing for the selection of the most robust and clinically relevant model. This framework enables a comprehensive evaluation of drift-aware learning systems for lifestyle disease modeling and supports anomaly reporting for improved decision-making in dynamic healthcare settings.

As described in Algorithm 1, the adaptive buffering approach is designed to handle evolving data distributions through drift detection mechanisms, ensuring the robustness of the model over time.

VII. DATASET DESCRIPTION AND PREPARATION

The dataset used in this experimentation is derived from [25] which is a nationally representative, cross-sectional dataset developed and maintained by the National Center for Health Statistics (NCHS), a division of the CDC in the United States. The original dataset between the year 2017-18 was split across multiple CSV files, each covering a different aspect of participant data, such as demographics, laboratory results, dietary intake, and medical questionnaires. The following are the preprocessing steps done in the process,

- 1) Loading 6 data sources: Demographics, Examination, Lab Results, Diet, Questionnaire, and Medical.
- 2) Selecting relevant features based on their significance to lifestyle disease modeling.
- 3) Merging all datasets using the common identifier column `SEQN`.
- 4) Handling categorical features via label encoding.
- 5) Estimating glucose levels from HbA1c using the formula: $Glucose = (HbA1c \times 28.7) - 46.7$.

- 6) Replacing missing values with column-wise medians.
- 7) Filtering invalid diabetes responses and creating a binary class label for diabetes diagnosis

The cumulated processed dataset contains 9,236 records and 26 features, with the class label being the Boolean value of being diabetic or not. The features selected based on relevance from the different survey files of the dataset are summarized in table 3.

A. CLASS BALANCING

Upon performing the Exploratory on the dataset and examining the class distribution as shown in figure 5, it is evident the class labels are not balanced, as the majority of the dataset consists of non-diabetic individuals. To balance this data, we have applied SMOTE (synthetic minority oversampling technique), which uses k-nearest neighbor to generate a synthetic minority sample equating to the number of the majority class samples. BMI (BMXBMI), age (RIDAGEYR), and glucose proxy (LBXGLU) show a moderate positive correlation with the diabetes label (DIQ010). Certain variables, like HDL cholesterol (LBDHDD) and physical activity (PAQ605, PAQ620), show negative correlation with diabetes, suggesting protective associations. Sociodemographic features such as education level (DMDEDUC2) and income (INDFMPIR) show mild correlations, indicating possible social determinants of diabetes.

B. DRIFT INDUCTION

To evaluate the robustness of concept drift detection and adaptation methods, three types of drift, abrupt, gradual, and recurring were synthetically injected into the balanced NHANES dataset. The target variable `DIQ010`, representing the binary class label, was manipulated to simulate drift

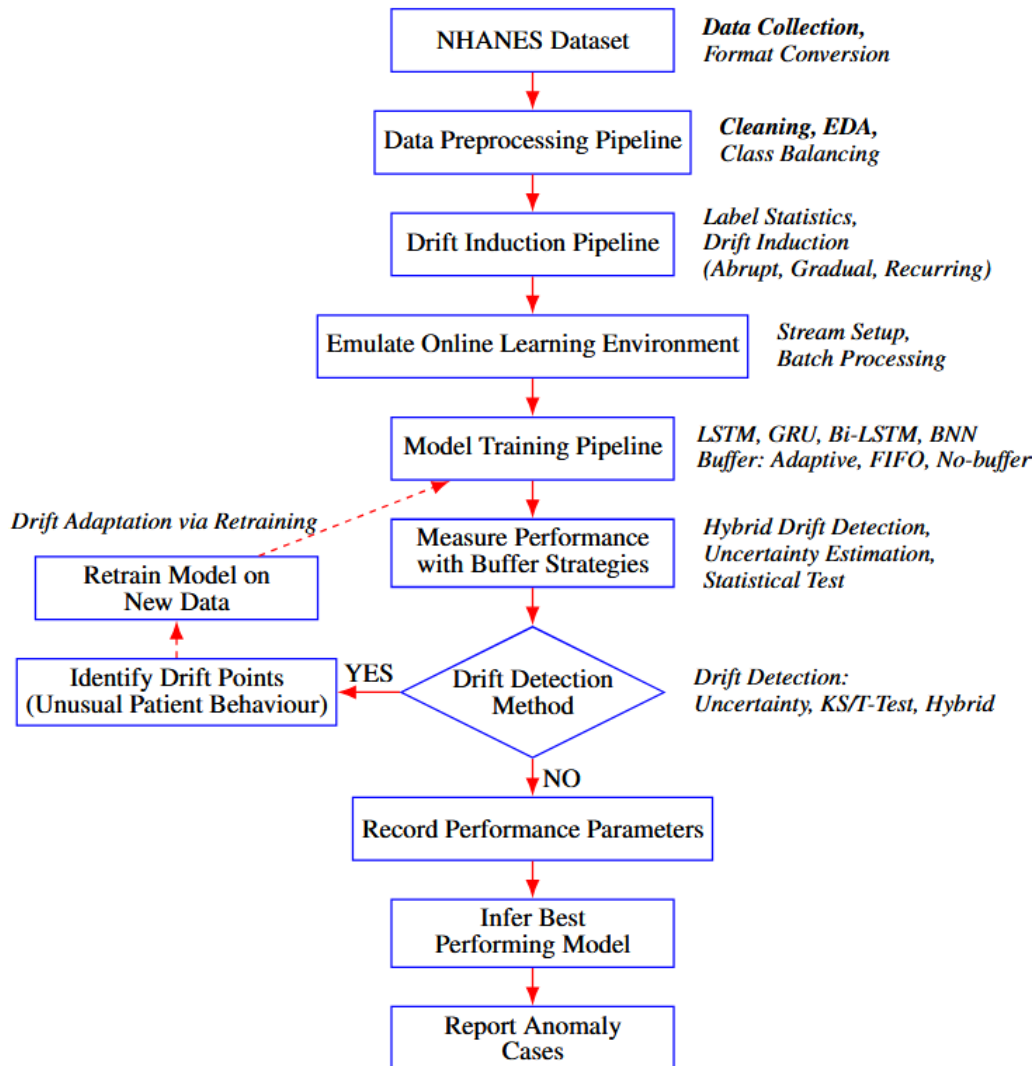


FIGURE 4. Proposed methodology: adaptive incremental learning pipeline for concept drift detection in lifestyle disease modeling.

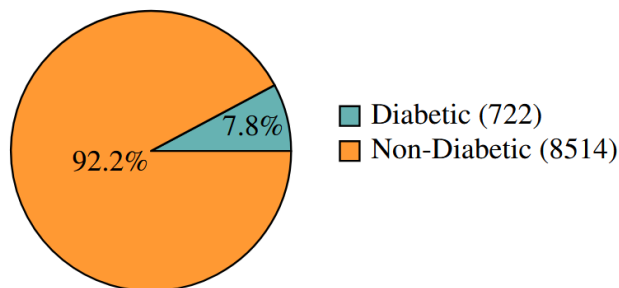


FIGURE 5. Class distribution of the original dataset.

scenarios. Label flipping was applied probabilistically to emulate real-world drift behavior while maintaining control over the drift magnitude and region. The following subsections describe the drift injection strategies implemented in this study. More realistic drift setup up would be

including the change in the distribution of each features (data drift), alongside the class label (concept drift) due to the measurement device failure or any equipment discrepancies or due to an environment anomalies that can happen over time.

1) ABRUPT DRIFT INJECTION

Abrupt drift simulates a sudden change in the data distribution at a specific point in time. To induce abrupt drift, a single contiguous region of the dataset is selected starting from a predefined drift index. Labels within this region are flipped with a fixed probability, referred to as the *drift magnitude*. A higher drift magnitude implies a more severe label transition.

Let s be the drift start index, w be the window size, and p be the drift magnitude. For each instance $i \in [s, s + w)$, the class label y_i is flipped to $1 - y_i$ with probability p .

Algorithm 1 Adaptive Buffering for Concept Drift Detection in Lifestyle Disease Prediction

```

1: Input: NHANES Dataset  $D = \{(x_t, y_t)\}_{t=1}^T$ 
2: Output: Drift detection and model adaptation
3: Perform Data Preprocessing:
4:   Clean the data: Remove noise and missing values
5:   Perform Exploratory Data Analysis (EDA)
6:   Balance the classes using techniques like SMOTE
7: Simulate Drift:
8:   Inject drift into dataset:
9:     For abrupt drift:  $y'_t \leftarrow y_t \oplus p$ , where  $p$  is the drift magnitude
10:    For gradual drift:  $y'_t \leftarrow f(p_t, y_t)$ , where  $p_t$  is the drift over time
11:    For recurring drift: Apply drift periodically at  $\{t_1, t_2, \dots\}$ 
12: Emulate Online Learning:
13:   Set up streaming environment:  $\{(x_t, y_t)\}_{t=1}^T$ 
14:   Divide data into mini-batches:  $\{B_1, B_2, \dots, B_n\}$ 
15: Model Training:
16:   Train models on mini-batches:  $f_\theta \leftarrow \text{Train}(B_i)$  for  $i = 1, \dots, n$ 
17:   Apply Buffer Strategies: Adaptive, FIFO, No-buffer
18: Measure Performance:
19:   Evaluate model with metrics: Accuracy, Precision, Recall, F1-Score, AUC
20:   For each batch, calculate performance:
21:     Performance =  $\frac{1}{n} \sum_{t=1}^n \mathbb{I}[y_t = \hat{y}_t]$ 
22: Detect Drift:
23: if Drift Detected then
24:   Identify drift points using uncertainty, KS-Test, Hybrid methods
25:   For uncertainty-based detection:
26:     Uncertainty( $x_t$ ) =  $\mathbb{E}[\hat{y}_t|x_t] - \mathbb{E}[\hat{y}_{t-1}|x_{t-1}]$ 
27:   For KS-Test or T-Test:
28:      $H_0 : \mu_t = \mu_{t-1}$  vs  $H_A : \mu_t \neq \mu_{t-1}$ 
29:   Update model on drift points:
30:      $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(f_{\theta_t}, B_{\text{drift}})$ 
31:   Re-train the model with drifted data:  $f_\theta \leftarrow \text{Train}(B_{\text{drift}})$ 
32: else
33:   Record performance parameters for future evaluation:
34:      $P_{\text{record}} = \{\text{Accuracy, Precision, Recall, F1-Score, } \dots\}$ 
35: end if
36: Output: Best performing model and drift report  $\mathcal{R}$ 

```

2) GRADUAL DRIFT INJECTION

Gradual drift emulates a scenario where the data distribution changes progressively over time. To simulate this, a wider window is chosen and a linearly or non-linearly increasing probability is used to flip the labels.

The drift probability at position i in the window is defined as:

$$P(i) = \begin{cases} \text{Linear:} & p \cdot \frac{i}{w} \\ \text{Quadratic:} & p \cdot \left(\frac{i}{w}\right)^2 \\ \text{Sigmoid:} & p \cdot \frac{1}{1 + e^{-z_i}}, \quad z_i \in [-6, 6] \end{cases}$$

where p is the maximum drift magnitude, and w is the size of the drift window.

This ensures a smooth transition from no drift to maximum drift across the window, more closely resembling real-world drift evolution.

3) RECURRING DRIFT INJECTION

Recurring drift captures patterns where the same or similar drift recurs at multiple intervals. This is typical in seasonal or cyclic systems. Drift regions are selected at multiple, pre-defined positions throughout the dataset, and label flipping is performed within each region using a constant or variable drift magnitude.

Let $B = \{b_1, b_2, \dots, b_n\}$ denote the set of batch indices where drift recurs. For each $b_j \in B$, a window of size w is defined, and within that window, each label is flipped with probability p_j , which may vary per recurrence.

The summary of the induced drift points for Abrupt, Gradual and Recurring drift are shown in table 4 with

TABLE 3. Description of the columns selected for experimentation from dataset.

Column Name	Description
SEQN	Unique participant identifier (used for merging datasets)
RIDAGEYR	Age in years
RIAGENDR	Gender (1 = Male, 2 = Female, label-encoded)
RIDRETH1	Race/Ethnicity (categorical, encoded)
DMDEDUC2	Education level (categorical, encoded)
INDFMPIR	Poverty-Income Ratio (economic indicator)
BMXWT	Body weight (kg)
BMXHT	Height (cm)
BMXBMI	Body Mass Index
BPXSY1	Systolic blood pressure
BPXDI1	Diastolic blood pressure
DR1TKCAL	Total daily caloric intake (kcal)
DR1TPROT	Daily protein intake (grams)
DR1TSUGR	Daily sugar intake (grams)
DR1TFIBE	Daily fiber intake (grams)
DR1TSFAT	Daily saturated fat intake (grams)
LBDHDD	HDL cholesterol (mg/dL)
LBXTC	Total cholesterol (mg/dL)
LBXTR	Triglycerides (mg/dL)
LBDLDL	LDL cholesterol (mg/dL)
LBXGLU	Estimated glucose from HbA1c (mg/dL)
SMQ020	Smoking status (1 = Yes, 2 = No, encoded)
PAQ605	Work-related physical activity (encoded)
PAQ620	Recreational physical activity (encoded)
ALQ101	Alcohol use in past 12 months (encoded)
DIQ010	Class label: Diabetes diagnosis (1 = Diabetic, 0 = Non-diabetic)

4 statistical tests performed to confirm the presence of drift, they are, KS-Test, Chi-Squared, T-Test, and Wasserstein Distance measures. The methodology employed to confirm the drift is if any two tests has p -value < 0.05 , then drift is flagged and confirmed.

TABLE 4. Drift detection summary using statistical tests.

Statistical Tests	KS-Test, Chi-Squared, T-Test, Wasserstein Distance
Logic to flag Drift	Any 2 tests with p -value $< 0.05 \Rightarrow$ Drift = <i>yes</i>
Drift Detection Results	
Drift Type	Drift Confirmed / Drift Absent
Abrupt Drift	14 / 318
Gradual Drift	13 / 319
Recurring Drift	13 / 319

VIII. MODEL DESCRIPTION

The four deep learning models implemented in this study are BiLSTM, BNN, GRU, and LSTM which serve as temporal learners for detecting lifestyle disease patterns under concept drift. The BiLSTM (Bidirectional LSTM) model leverages both forward and backward temporal dependencies, enhancing contextual learning by processing input sequences from both directions. It uses dropout regularization and a fully connected layer followed by a sigmoid activation to produce probabilistic outputs. The BNN is based on an LSTM architecture but distinguishes itself by incorporating Monte Carlo (MC) dropout during

inference, enabling it to estimate prediction uncertainty. This is particularly beneficial in drift-prone environments where decision confidence varies. The GRU employs a Gated Recurrent Unit, which is computationally efficient compared to LSTMs and captures sequential dependencies while using fewer parameters. Lastly, the LSTM uses a standard unidirectional LSTM layer to model long-term dependencies in the input data. All models incorporate dropout and are designed to output a binary prediction indicating the presence or absence of diabetes, serving as core learners in the adaptive buffering framework. The descriptions of each of the models are summarized in table 5.

IX. EXPERIMENTAL SETUP AND EVALUATION CRITERIA

The experimentation is conducted on an Acer Nitro machine with a Ryzen 7 and a CUDA-enabled Nvidia GPU with graphics 3050 with 6GB graphics. To assess and validate the presence of concept drift in the context of lifestyle disease modeling, we employed three complementary strategies for drift point identification. Each strategy is designed to capture different dimensions of distributional and predictive shifts in the data stream.

1) Statistical Test-Based Detection (KS-Test and T-Test)

This method relies purely on statistical hypothesis testing. The Kolmogorov-Smirnov (KS) test is used to detect changes in the empirical distributions of key

TABLE 5. Summary of deep learning architectures used for drift-aware learning.

Model	Core Layer	Directionality	Dropout	Uncertainty Estimation
BiLSTM Learner	Bidirectional LSTM (nn . LSTM)	Bidirectional	Yes ($p = 0.3$)	No
BNN Learner	LSTM + MC Dropout	Unidirectional	Yes ($p = 0.3$)	Yes (MC sampling)
GRU Learner	GRU (nn . GRU)	Unidirectional	Yes ($p = 0.3$)	No
LSTM Learner	LSTM (nn . LSTM)	Unidirectional	Yes ($p = 0.3$)	No

features, while the two-sample T-test evaluates shifts in feature means between two temporal segments (typically pre-drift and post-drift batches). A statistically significant change in either test quantified via low p-values and large test statistics—indicates a potential drift point.

2) **Accuracy Dip Combined with Statistical Evidence**

In this hybrid method, drift points are flagged by tracking sudden declines in the model’s prediction accuracy over time. When a notable drop in accuracy is observed (relative to a rolling baseline), statistical tests (KS or T-Test) are applied to the surrounding batches to confirm distributional change. This approach combines model-centric performance degradation with data-centric evidence to avoid false positives and improve detection reliability.

3) **Uncertainty-Based Drift Detection**

This method leverages model uncertainty as a signal for drift. Using Monte Carlo dropout. A spike in epistemic uncertainty indicates that the model is encountering data it is less confident about, often due to a shift in the underlying data distribution. If the uncertainty increase exceeds a threshold, a drift is declared. This approach is particularly useful in dynamic settings where distributional change is gradual or subtle.

X. RESULTS AND DISCUSSION

This section discusses the performance of deep learning models (Bi-LSTM, BNN, LSTM, and GRU) across three drift detection strategies using adaptive, FIFO, and no-buffer incremental learning frameworks. The evaluation metrics include accuracy, precision, recall, F1-score, AUC, and the number of detected drift points, serving to quantify both predictive effectiveness and adaptability under dynamic clinical data streams. The experimentation is done in three drift detection verticals; they are,

- 1) **Hybrid Drift Detection:** Combines error rate drops with statistical validation.
 - 2) **Statistical Test-Based Detection:** Uses KS-Test and T-Test to detect distributional shifts.
 - 3) **Uncertainty-Based Detection:** Leverages predictive variance from Bayesian models for drift identification.
- The subsections below examine each of these vertical experimentation findings in detail.

A. HYBRID DRIFT DETECTION

The performance of 4 Deep learning models(Bi-LSTM, BNN, LSTM, and GRU) with 3 buffer strategies(Adaptive, FIFO and No-Buffer) with 3 Drift types(Abrupt, Gradual and Recurring) with Hybrid drift detection mechanism is summarized in the table 10. In the case of Abrupt Drift, the Adaptive Buffer Strategy yields BiLSTM detecting 4 drift points(figure 6) with 0.73 accuracy, while BNN (figure 7 performs slightly worse, detecting 2 drift points with 0.73 accuracy. GRU and LSTM have the best runtime performance, but GRU (figure 9) detects no drift points, while LSTM (figure 8) detects 2 drift points with similar accuracy. FIFO Buffer Strategy leads to BiLSTM detecting 3 drift points with 0.73 accuracy, and BNN maintaining accuracy but with a lower drift detection count. GRU and LSTM perform better in runtime efficiency, but BiLSTM shows the highest drift detection with slower runtime. The No Buffer Strategy exhibits better efficiency for all models, with BiLSTM detecting 3 drift points and performing with 0.73 accuracy. BiLSTM incurs the highest runtime in all buffer strategies, especially in FIFO (32.85 seconds) and No Buffer (30.01 seconds). The BNN model exhibits consistently high runtimes, particularly in Adaptive Buffer (23.79 seconds), indicating its computational cost for Bayesian inference. GRU and LSTM are faster and computationally efficient, with runtimes ranging between 15.09 and 24.29 seconds, depending on the drift type and buffer strategy.

For Gradual Drift, BiLSTM detects 0 drift points with 0.77 accuracy in Adaptive Buffer, while BNN performs better with 1 drift point detected at 0.77 accuracy. GRU and LSTM show better drift detection with 1 drift point each, achieving 0.77 accuracy and 0.84 AUC in Adaptive Buffer. FIFO Buffer Strategy again increases complexity, with BiLSTM detecting 1 drift point and BNN detecting 1 with 0.77 accuracy. LSTM and GRU show strong results with good accuracy. No Buffer Strategy maintains consistent performance across all models with 0.77 accuracy but achieves faster runtimes compared to the buffer strategies. BiLSTM experiences the longest runtimes in FIFO (49.1 seconds) and No Buffer (32.68 seconds), while BNN shows higher runtimes in both buffer strategies (233.02 seconds). GRU and LSTM have better runtime efficiency, particularly in the No Buffer Strategy, where GRU takes 22.92 seconds and LSTM takes 25.92 seconds. The Adaptive Buffer Strategy results in

moderate runtimes, with BiLSTM at 22.45 seconds and GRU at 15.28 seconds.

In the case of Recurring Drift, BiLSTM detects 2 drift points with 0.80 accuracy and 0.85 AUC in Adaptive Buffer, while BNN performs similarly with 1 drift point detected. GRU shows high performance with 1 drift point and 0.80 accuracy in Adaptive Buffer. FIFO Buffer Strategy increases complexity, with BiLSTM detecting 2 drift points and achieving 0.80 accuracy. GRU and LSTM show strong performance, with LSTM achieving 0.80 accuracy across all buffer types. No Buffer Strategy performs efficiently with BiLSTM detecting 2 drift points and performing with 0.80 accuracy in 33.37 seconds. BiLSTM and BNN continue to show high runtime costs in the FIFO Buffer Strategy, with BiLSTM taking 66.09 seconds and BNN taking 246.15 seconds. GRU and LSTM perform with higher efficiency in the FIFO Buffer Strategy, with runtimes of 47.06 seconds for GRU and 56.12 seconds for LSTM. In No Buffer, BiLSTM takes 33.37 seconds, while GRU and LSTM have faster runtimes of 22.06 seconds and 20.28 seconds, respectively.

Across all the types of drift, the Adaptive Buffer Strategy and FIFO Buffer Strategy consistently provide better drift detection and accuracy, particularly for gradual and recurring drifts. The No Buffer Strategy results in lower performance, especially for abrupt drift, where drift detection and accuracy are compromised. BiLSTM and BNN are the most robust models, showing stable performance across all drift types and strategies, while GRU tends to underperform, especially in detecting drifts.

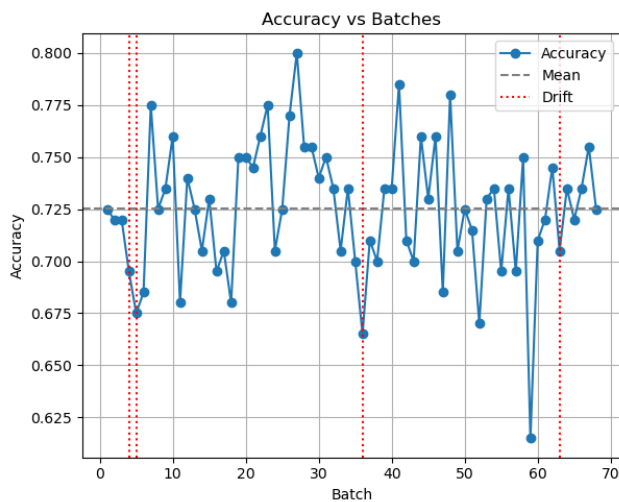


FIGURE 6. Performance of BiLSTM with hybrid drift detection using adaptive buffering: Accuracy trend(abrupt).

B. STATISTICAL TEST-BASED DRIFT DETECTION

The performance of 4 Deep learning models(Bi-LSTM, BNN, LSTM, and GRU) with 3 buffer strategies(Adaptive, FIFO and No-Buffer) with 3 Drift types(Abrupt, Gradual

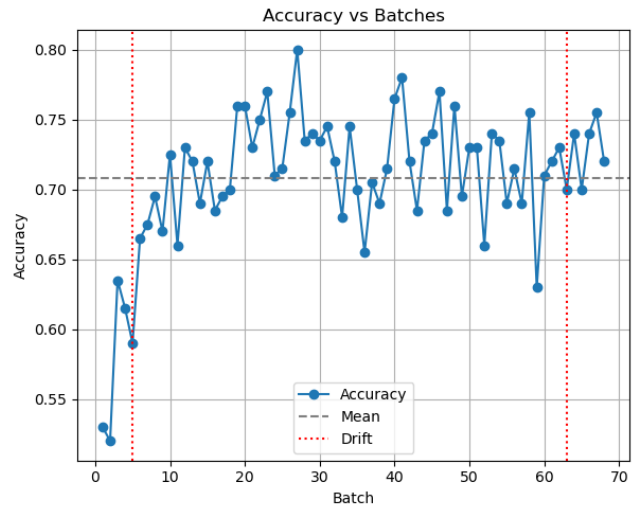


FIGURE 7. Performance of BNN with hybrid drift detection using adaptive buffering: Accuracy trend(abrupt).

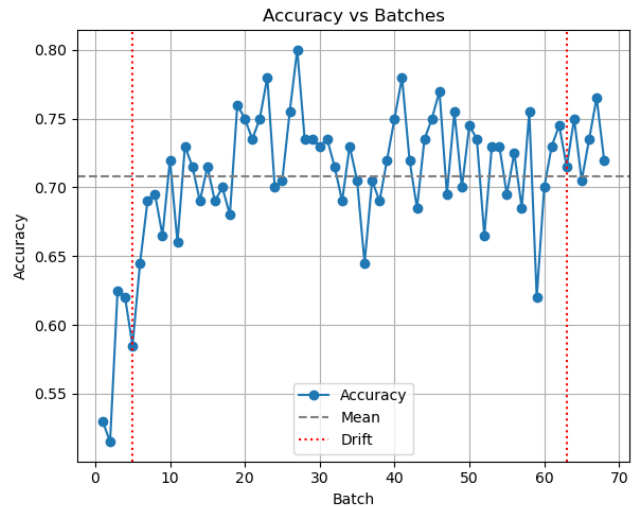


FIGURE 8. Performance of LSTM with hybrid drift detection using adaptive buffering: Accuracy trend(abrupt).

and Recurring) with External drift detection using KS-test is summarized in the table 11.

In the Abrupt Drift scenario, the Adaptive Buffer Strategy yields the best results, with BiLSTM (figure 10) detecting 18 drift points and achieving an accuracy of 0.73, maintaining good precision and recall. The GRU (figure 13) model also performs well with 18 drift points detected and slightly higher accuracy (0.74). The BNN (figure 12) model detects fewer drift points (15) and maintains similar accuracy to BiLSTM (0.73). LSTM (figure 11) performs slightly worse, detecting 13 drift points with an accuracy of 0.73. In the FIFO Buffer Strategy, drift detection is lower, with BiLSTM detecting 12 drift points and an accuracy of 0.72. The GRU model performs similarly, detecting 12 drift points and maintaining 0.73 accuracy, while LSTM detects 15 drift points with

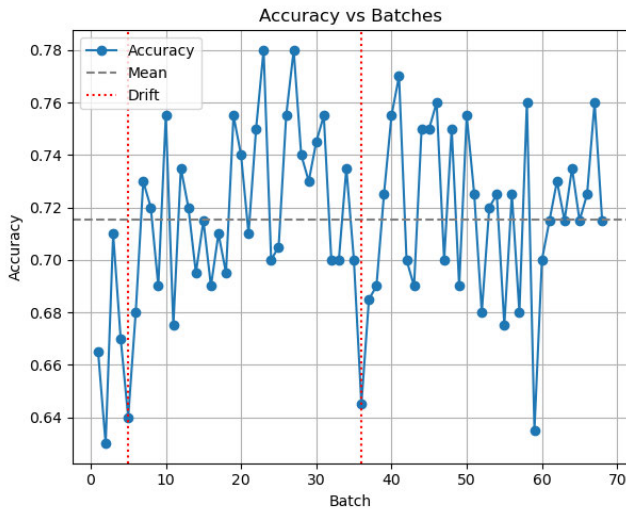


FIGURE 9. Performance of GRU with hybrid drift detection using adaptive buffering: Accuracy trend(abrupt).

0.73 accuracy. For the No Buffer Strategy, BiLSTM performs the best, detecting 22 drift points with 0.81 accuracy, while GRU and LSTM perform similarly but with fewer drift points detected. BNN incurs the highest runtime, particularly in the Adaptive Buffer strategy (27.67 seconds), while GRU and LSTM have faster runtimes of 23.25 seconds and 19.96 seconds, respectively. The FIFO Buffer strategy leads to longer runtimes, especially for BiLSTM (35.28 seconds). No Buffer Strategy offers the most computational efficiency, with BiLSTM taking 22 seconds and GRU and LSTM taking less than 15 seconds.

For Gradual Drift, the Adaptive Buffer Strategy again delivers good results. BiLSTM detects 15 drift points with an accuracy of 0.78, while BNN detects the same number of drift points with accuracy remaining at 0.78. GRU detects 18 drift points but performs slightly worse in accuracy (0.76). LSTM shows similar performance to the other models, detecting 16 drift points and achieving 0.78 accuracy. In the FIFO Buffer Strategy, BiLSTM and BNN both achieve 0.78 accuracy, with BiLSTM detecting 15 drift points and BNN detecting 18 drift points. The GRU model, in FIFO, detects fewer drift points (13) but maintains accuracy at 0.78. LSTM also performs similarly with 15 drift points detected and 0.78 accuracy. In the No Buffer Strategy, all models perform well with an accuracy of 0.80. BiLSTM, BNN, and LSTM perform similarly with 14 to 16 drift points detected, while GRU detects 17 drift points but maintains 0.80 accuracy. GRU shows the shortest runtime of 11.89 seconds in the Adaptive Buffer Strategy, while BiLSTM takes 22.45 seconds. The FIFO Buffer Strategy increases runtimes significantly, with BiLSTM at 49.1 seconds and BNN at 28.14 seconds. The No Buffer Strategy provides a good trade-off, with GRU performing in 14-15 seconds and LSTM taking 20.39 seconds.

For Recurring Drift, the Adaptive Buffer Strategy provides the highest performance across all models, with BiLSTM detecting 17 drift points and achieving 0.80 accuracy. The BNN model performs similarly, detecting 18 drift points and achieving the same accuracy of 0.80. GRU and LSTM also show strong performance, detecting 14 to 16 drift points with 0.80 accuracy. In the FIFO Buffer Strategy, BiLSTM detects 14 drift points with 0.80 accuracy, while BNN and LSTM perform similarly, detecting 15 drift points and maintaining 0.81 accuracy. GRU performs slightly worse, with 13 drift points detected and 0.80 accuracy. The No Buffer Strategy yields comparable performance with 0.80 accuracy across all models. BiLSTM, BNN, and LSTM detect 14 to 16 drift points, while GRU detects 17 drift points with slightly lower performance in precision and recall. BiLSTM incurs the highest runtime in FIFO Buffer (84.7 seconds), while BNN and LSTM take 55.97 and 56.12 seconds, respectively. GRU performs better, with 47.06 seconds in FIFO. In the No Buffer Strategy, BiLSTM and BNN exhibit good runtime efficiency (55.59 and 35.4 seconds), while GRU and LSTM are faster, taking 29.48 and 31.35 seconds, respectively.

In case of Statistical test based drift detection mechanism, GRU and LSTM are the most computationally efficient models across the drift types, providing a reasonable balance between accuracy and runtime. The No Buffer Strategy offers the best runtime efficiency, especially with GRU and LSTM, but at the expense of some drift detection accuracy in Gradual and Recurring drifts. BiLSTM and BNN offer the highest accuracy, especially for more complex drift types like Abrupt and Recurring but incur significantly higher runtimes, particularly with the FIFO Buffer strategy. FIFO Buffer Strategy tends to increase runtime considerably, especially with BiLSTM and BNN, making it less suitable for real-time applications but beneficial in scenarios where accuracy is prioritized over speed. The Adaptive Buffer Strategy provides a balance but still requires moderate runtime, with BiLSTM performing the best in accuracy and drift detection.

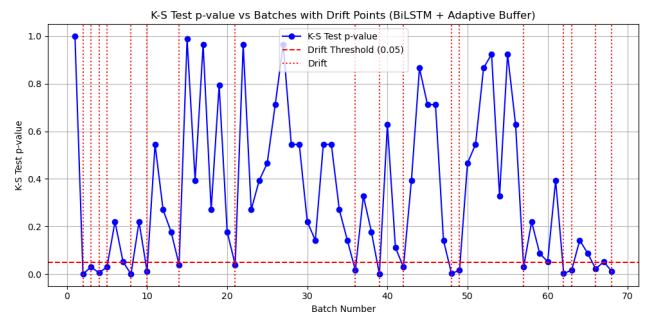


FIGURE 10. Performance of BiLSTM with KS-test based drift detection using adaptive buffering: Accuracy trend(abrupt).

C. UNCERTAINTY BASED DRIFT DETECTION

The performance of 4 Deep learning models(Bi-LSTM, BNN, LSTM, and GRU) with 3 buffer strategies(Adaptive,

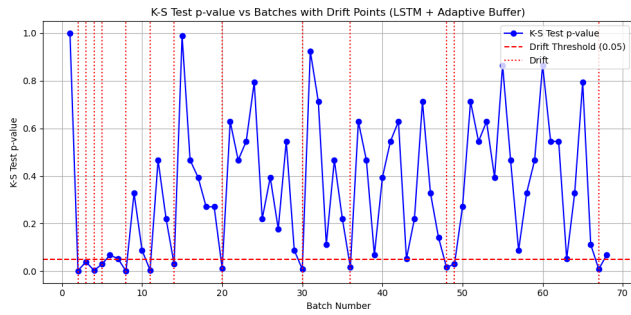


FIGURE 11. Performance of LSTM with KS-test based drift detection using adaptive buffering: Accuracy trend(abrupt).

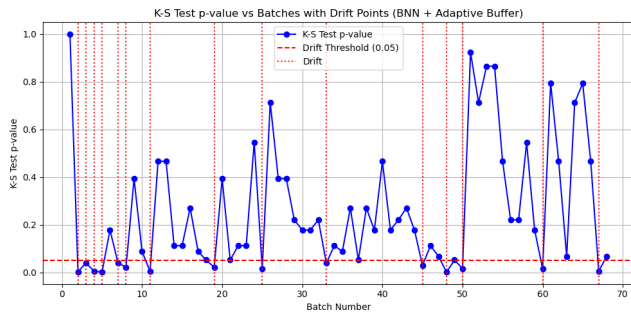


FIGURE 12. Performance of BNN with KS-test based drift detection using adaptive buffering: Accuracy trend(abrupt).

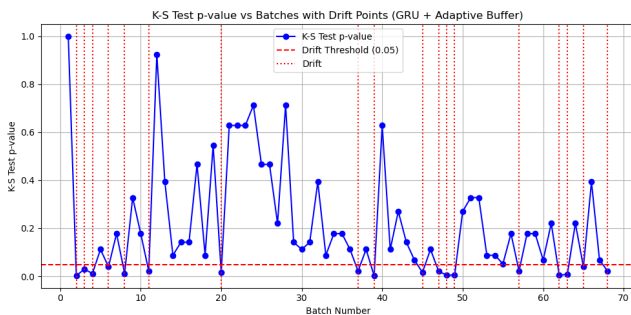


FIGURE 13. Performance of GRU with KS-test based drift detection using adaptive buffering: Accuracy trend(abrupt).

FIFO and No-Buffer) with 3 Drift types(Abrupt, Gradual and Recurring) with uncertainty based drift detection is summarized in the table 12.

In the Abrupt Drift scenario, the Adaptive Buffer Strategy shows BiLSTM (figure 14) detecting 14 drift points with an accuracy of 0.73, while GRU (figure 17) performs slightly better, detecting 19 drift points and achieving an accuracy of 0.72. BNN (figure 15) performs worse, detecting 9 drift points and having an accuracy of 0.7, with its computational complexity reflected in a significantly higher runtime of 233.01 seconds. LSTM (figure 16) performs similarly to BiLSTM, detecting 21 drift points with 0.73 accuracy, and the runtime is moderate at 24.48 seconds. In the FIFO Buffer Strategy, BiLSTM detects 17 drift points and maintains an

accuracy of 0.73 but has a higher runtime of 50.42 seconds, reflecting the added complexity of maintaining a FIFO buffer. BNN continues to show high runtime (363.45 seconds), confirming its computationally expensive nature. GRU and LSTM show lower runtime, with GRU taking 26.21 seconds and LSTM 37.79 seconds, indicating that these models are more efficient in handling drift detection with a buffer. In the No Buffer Strategy, BiLSTM detects 18 drift points with 0.73 accuracy in 36.26 seconds, while GRU and LSTM show faster runtime (17.95 and 26.92 seconds, respectively), with GRU showing good performance despite fewer drift points detected.

For Gradual Drift, the Adaptive Buffer Strategy provides good performance with BiLSTM detecting 17 drift points with 0.78 accuracy and 0.76 F1-score. BNN, however, detects fewer drift points (9) and has lower accuracy (0.69) and runtime (233.02 seconds), indicating that Bayesian models are less efficient for gradual drift detection. GRU detects 15 drift points with 0.77 accuracy, while LSTM shows slightly better performance with 20 drift points detected and 0.77 accuracy. The FIFO Buffer Strategy again results in longer runtime, especially for BiLSTM with 60.73 seconds, compared to BNN and GRU, which have 307.21 seconds and 358.51 seconds runtimes, respectively. LSTM performs similarly to BiLSTM, with 20 drift points detected and 0.78 accuracy, but with a more reasonable runtime (49.89 seconds). The No Buffer Strategy shows better efficiency, with BiLSTM detecting 17 drift points and achieving 0.77 accuracy in 33.93 seconds, while GRU and LSTM show good performance with runtime of 17.95 and 26.92 seconds, respectively.

For Recurring Drift, the Adaptive Buffer Strategy results in BiLSTM detecting 19 drift points with 0.80 accuracy and 0.86 AUC in 33.45 seconds, while BNN again shows lower accuracy (0.69) and runtime (233.01 seconds). GRU performs similarly, detecting 15 drift points with 0.80 accuracy and 0.85 AUC, and LSTM achieves 0.80 accuracy with 21 drift points detected. The FIFO Buffer Strategy increases the runtime significantly, especially for BiLSTM (66.09 seconds), BNN (246.15 seconds), and GRU (294.67 seconds). LSTM continues to perform well with 20 drift points detected and 0.81 accuracy in 46.95 seconds. The No Buffer Strategy results in 0.80 accuracy across all models, with BiLSTM detecting 17 drift points in 36.18 seconds. GRU and LSTM continue to show fast runtime, with GRU taking 20.92 seconds and LSTM at 25.75 seconds.

The No Buffer Strategy offers the lowest computational overhead and quickest runtime, making it ideal for applications where speed is crucial, though it may sacrifice some accuracy in drift detection. The Adaptive Buffer Strategy provides a good balance of drift detection and runtime, with GRU and LSTM being more computationally efficient compared to BiLSTM and BNN. BNN, while accurate, is computationally expensive and should be used in environments where high accuracy justifies the additional time complexity. The FIFO Buffer Strategy, though useful for improved drift detection,

introduces significant computational overhead, especially for BiLSTM and BNN, making it less practical for real-time systems.

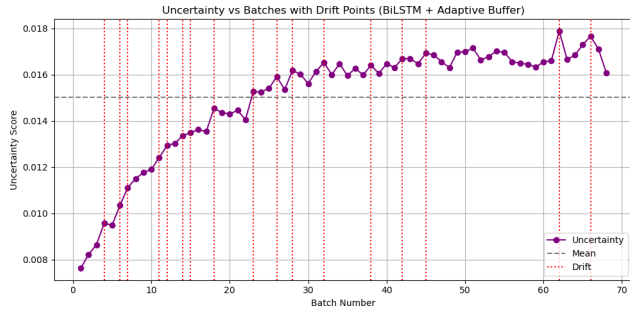


FIGURE 14. Performance of BiLSTM with Uncertainty based Drift detection using adaptive buffering: Accuracy trend(abrupt).

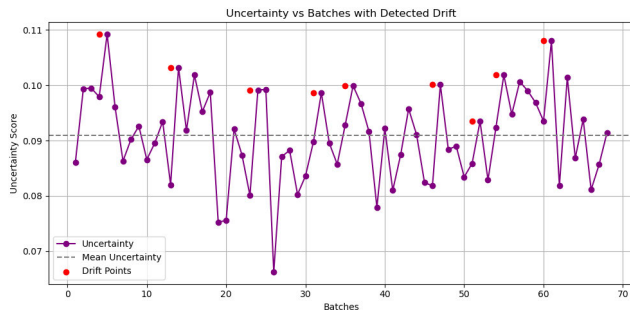


FIGURE 15. Performance of BNN with uncertainty based drift detection using adaptive buffering: Accuracy trend(abrupt).

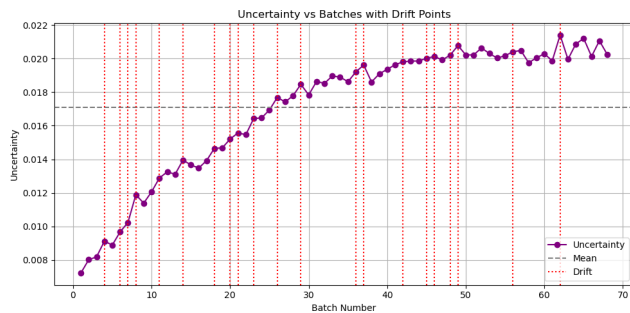


FIGURE 16. Performance of LSTM with Uncertainty based drift detection using adaptive buffering: Accuracy trend(abrupt).

D. ABLATION STUDY ON DRIFT DETECTION METHODS

This section presents the ablation studies conducted for two different drift detection techniques used in our experiments: *Hybrid Drift Detection* and *Uncertainty-Based Drift Detection*. These studies aim to isolate and evaluate the impact of different components of the drift detection methods on overall performance, accuracy, and computational efficiency.

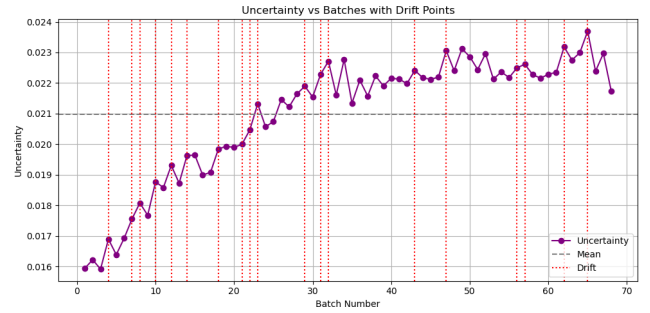


FIGURE 17. Performance of GRU with uncertainty based drift detection using adaptive buffering: Accuracy trend(abrupt).

1) ABLATION STUDY ON HYBRID DRIFT DETECTION WITH ADAPTIVE BUFFERING

The first study focuses on the Hybrid Drift Detection method combined with adaptive buffering. The goal is to evaluate how different aspects of the drift detection process, such as drift-triggered buffering, retraining, and drift detection mechanisms (KS-test, error-rate), affect the model's ability to detect and adapt to drift.

Description of Ablations: The following are the ablations taken for study and its interpretation as summarized in the table 6,

- 1) B1 (Base Line): All features are enabled: drift-triggered buffering, retraining, and the use of the KS-test for detecting drift.
- 2) B2: Drift-triggered buffering is disabled, meaning drift is added to the buffer only when detected
- 3) B3: Buffer retraining is disabled, and the model does not retrain once drift is detected
- 4) B4: The KS-test is disabled, and the model uses only error-rate methods to detect drift

The B1 configuration is the reference point for comparison, and we observe how disabling each component affects performance.

Table 13 summarizes the findings of the 4 ablations for hybrid drift detection for the adaptive buffer strategy with Deep learners. The following are the key insights learned through this ablation study on each drift type.

- **Abrupt Drift:** In the case of Abrupt Drift, the Base Line (B1) configuration, with all components enabled, yielded consistent results across all models, including BiLSTM, BNN, GRU, and LSTM, with accuracy around 0.73 and AUC at 0.78. The runtime varied, with BiLSTM taking 24.29 seconds, and LSTM being slightly faster at 15.09 seconds. When Drift-triggered Buffering was disabled in B2, the runtime increased significantly (e.g., BiLSTM took 145.01 seconds), but the Accuracy and AUC improved to 0.82 and 0.87, respectively. Disabling Buffer Retraining (B3) resulted in a moderate decrease in accuracy (0.81) but significantly reduced the runtime (e.g., BiLSTM at 15.7 seconds). Lastly, in B4, when KS-test was replaced with the error-rate method, the models exhibited consistent performance similar to

TABLE 6. Ablation study on hybrid drift detection with adaptive buffering.

Ablation ID	Description
B1	Base Line (All Enabled)
B2	Disable Drift-triggered Buffering, Add to Buffer Only When Drift is Detected
B3	Disable Buffer Retraining, No Retraining
B4	Disable KS-test, Use Only Error-rate for Drift Detection

the baseline, with Accuracy around 0.81 and AUC at 0.86, while the runtime decreased in comparison to B2.

- **Gradual Drift:** For Gradual Drift, the Base Line (B1) configuration again provided strong performance, with Accuracy ranging from 0.77 to 0.78 and AUC at 0.84. The runtime for models like GRU and LSTM ranged between 15.28 and 16.12 seconds. In B2, where Drift-triggered Buffering was disabled, the Accuracy improved slightly to 0.79, and AUC reached 0.85, but the runtime increased significantly, with BiLSTM taking 77.42 seconds. Disabling Buffer Retraining (B3) led to a minor performance dip, with accuracy around 0.78 for BiLSTM, and the runtime dropped to 17.85 seconds. In B4, when KS-test was excluded, Accuracy decreased slightly to 0.79, and AUC maintained at 0.85, with runtime for BiLSTM decreasing to 23.87 seconds.
- **Recurring Drift:** For Recurring Drift, the Base Line (B1) configuration provided high Accuracy at 0.8 and AUC at 0.85. Models like BiLSTM and GRU performed well, with runtime around 25.65 seconds for BiLSTM and 15.51 seconds for LSTM. In B2, where Drift-triggered Buffering was disabled, the performance decreased, particularly for BiLSTM, with Accuracy dropping to 0.74, but the AUC remained at 0.78. This configuration resulted in a significant runtime increase (e.g., BiLSTM at 79.67 seconds). B3, which disabled Buffer Retraining, showed reduced runtime (e.g., BiLSTM at 28.15 seconds), but the Accuracy for BiLSTM was still lower (0.73) than in B1. Finally, in B4, where KS-test was excluded, the models exhibited a slight reduction in Accuracy and AUC (e.g., BiLSTM at 0.73 and 0.77, respectively), but still maintained reasonable performance with runtime ranging from 24.29 to 44.2 seconds.

This ablation study on Hybrid Drift Detection shows that Drift-triggered Buffering (B2) significantly improves Accuracy but incurs a substantial computational cost, increasing runtime. On the other hand, Buffer Retraining (B3) reduces computational time but leads to a slight drop in performance. Excluding the KS-test (B4) results in a modest reduction in accuracy and AUC, though it still offers a trade-off between performance and runtime efficiency.

E. ABLATION STUDY ON UNCERTAINTY-BASED DRIFT DETECTION

The second study focuses on the Uncertainty-Based Drift Detection method, which relies on Monte Carlo (MC) Dropout for uncertainty estimation. The ablation study

examines how different configurations of MC Dropout, buffer updates, and retraining affect the drift detection process.

Description of Ablations: The following are the ablations for Uncertainty based drift detection with adaptive buffers taken for study and its interpretation as summarized in the table 7,

- 1) U1 (Base Line): This configuration serves as the base, with an uncertainty threshold of 20 samples. Buffer updates and retraining are performed only when drift is detected, ensuring that the model adapts when drift occurs but without constant retraining or updates.
- 2) U2 (Continuous Updates and Retraining): In this setup, there is no uncertainty threshold (disabled), and both buffer updates and retraining occur every batch. This configuration tests the impact of continuous updates and retraining, which may result in high computational costs without necessarily improving the detection of drift.
- 3) U3 (Reduced Sensitivity): By setting the uncertainty threshold to 1 sample, this configuration lowers the sensitivity of drift detection. The buffer updates and retraining still occur on drift detection, but the reduced threshold may cause the model to miss subtle drifts, leading to lower performance in detecting gradual changes.
- 4) U4 (No Buffer Update and No Retraining): This configuration uses an uncertainty threshold of 20 samples for drift detection but disables both buffer updates and retraining. Without any adaptation mechanisms, the model may struggle to adjust to changing data distributions, resulting in decreased drift detection effectiveness.
- 5) U5 (Frequent Buffer Updates): Here, the uncertainty threshold remains at 20 samples, with buffer updates occurring every batch. Retraining is only performed on drift detection. This configuration examines the impact of frequent buffer updates on performance while restricting retraining to drift detection events.

Table 14 summarizes the findings of the 4 ablations for uncertainty based drift detection for adaptive buffer strategy with Deep learners. The following are the key insights learned through this ablation study on each drift type using uncertainty based drift detection method.

- **Abrupt Drift:** In the Abrupt Drift scenario, the U1 configuration (Uncertainty threshold with 20 samples, buffer update and retraining on drift) showed relatively consistent performance across models. BiLSTM and LSTM both achieved Accuracy values of 0.73, with

TABLE 7. Ablation study on uncertainty-based drift detection.

Ablation ID	Description
U1	Uncertainty threshold with 20 samples, buffer update and retraining on drift detection.
U2	No uncertainty threshold, buffer update and retraining every batch.
U3	Uncertainty threshold with 1 sample, buffer update and retraining on drift detection.
U4	Uncertainty threshold with 20 samples, no buffer update, no retraining.
U5	Uncertainty threshold with 20 samples, buffer update every batch, retraining on drift detection.

AUC at 0.78 and moderate runtime around 28.5 and 24.48 seconds, respectively. The BNN model, however, performed poorly with Accuracy dropping to 0.7 and a significantly higher runtime of 233.01 seconds. The U2 configuration, where retraining occurred on every batch, yielded significant improvements in Accuracy (around 0.82) and AUC (0.86) across all models, with BiLSTM showing the highest runtime of 86.93 seconds. The U3 configuration, with the uncertainty threshold and buffer update only on drift, displayed stable performance with Accuracy around 0.81 and lower runtime for GRU (13.6 seconds) and LSTM (11.59 seconds). In U4, when buffer retraining was disabled, performance remained close to U2, but runtime was significantly reduced (e.g., BiLSTM at 39.57 seconds). Finally, in U5, where both the buffer update occurred every batch and retraining was performed on drift, the Accuracy remained around 0.81, but the runtime increased slightly for BiLSTM to 63.84 seconds.

- **Gradual Drift:** For Gradual Drift, U1 exhibited consistent performance across models, with BiLSTM achieving Accuracy of 0.78 and AUC at 0.84, with moderate runtime ranging from 22.45 to 25.84 seconds. U2 (buffer update on drift) showed improved Accuracy (ranging from 0.79 to 0.80) and AUC (0.85) across all models, but the runtime increased substantially for BiLSTM (108 seconds). U3 saw a performance drop in Accuracy for BNN to 0.79, but the runtime remained relatively low for GRU (8.26 seconds) and LSTM (8.32 seconds). In U4, the performance was stable with Accuracy remaining around 0.79, but the runtime was notably reduced (e.g., BiLSTM at 24.98 seconds). In U5, which had both batch-wise updates and retraining, the models showed slightly decreased Accuracy (e.g., 0.75 for BiLSTM) but maintained AUC at 0.85, with runtime ranging from 35.46 seconds for BiLSTM to 18.18 seconds for GRU.
- **Recurring Drift:** In the Recurring Drift scenario, U1 exhibited high Accuracy at 0.8 and AUC at 0.86, with GRU and LSTM showing reasonable runtime performance (e.g., 24.12 and 27.06 seconds, respectively). The U2 configuration saw a significant decrease in Accuracy (around 0.74) and AUC at 0.78, but the performance was stable across models with runtime varying from 79.67 seconds for BiLSTM to 56.15 seconds for LSTM. In U3, performance was consistently maintained, with Accuracy for BiLSTM at 0.73 and AUC at 0.77, and

runtime remaining low (e.g., 15.42 seconds for GRU). The U4 configuration, with uncertainty-based updates on drift and retraining, showed similar performance to U3, maintaining Accuracy at 0.73 for BiLSTM and AUC at 0.77, with runtime around 31.96 seconds for BiLSTM. Finally, in U5, when both batch updates and retraining were enabled, performance was slightly lower (e.g., Accuracy for BiLSTM at 0.71) but still acceptable, with runtime being the highest at 43.2 seconds for BiLSTM.

This ablation study highlights the effects of different configurations in Uncertainty-based Drift Detection. The best overall performance was observed in U2, where both the buffer update occurred on drift and retraining was performed, leading to high Accuracy and AUC for all models. However, this came at the cost of increased runtime, particularly for BiLSTM. The U3 configuration, which utilized uncertainty-based updates only on drift, achieved good performance with reduced runtime, especially for models like GRU and LSTM. Finally, the U4 and U5 configurations provided trade-offs between performance and computational cost, with U4 being the most balanced option in terms of runtime and performance.

F. DRIFT MAGNITUDE STUDY

In this section, we analyze the impact of varying drift magnitudes (20%, 50%, 80%, and 100% flips) on different drift detection methods: Hybrid Approach, Uncertainty-based Drift Detection, and KS-Test-based Drift Detection. The goal is to observe how different models respond to the severity of the induced drift and how these adjustments affect performance metrics, including Accuracy, Precision, Recall, F1-Score, AUC, and CPU Time.

The drift is introduced by flipping a certain percentage of the dataset's labels, and the models are evaluated under these varying conditions. By comparing these performance metrics across different drift magnitudes, we can gauge the robustness and efficiency of each detection method in handling abrupt, gradual, and recurring concept drift types. Table 15 summarizes the findings of the drift magnitude study against the various drift detection methods using adaptive buffering strategy.

- **Abrupt Drift:** For Abrupt Drift, the Hybrid Approach demonstrated that BiLSTM is the most efficient model, achieving the highest Accuracy of 0.94 for a 20% flip with a moderate CPU Time of 5.5 seconds. As the drift magnitude increased, the Accuracy decreased, reaching 0.81 for a 100% flip, but still maintaining

solid performance. The GRU and LSTM models also exhibited similar patterns, where Accuracy gradually dropped, but their Performance remained competitive, with Accuracy ranging from 0.74 to 0.75 in the worst case. These results highlight that the Hybrid Approach retains high performance in handling Abrupt Drift, with BiLSTM showing the most consistent and reliable results across all levels of drift. For Abrupt Drift, the Uncertainty-based Drift Detection method showed that BiLSTM achieved solid performance, maintaining Accuracy at 0.87 for a 20% flip and dropping to 0.80 for a 100% flip. However, BNN demonstrated a substantial performance drop in Accuracy, starting from 0.9 at a 20% flip and decreasing to 0.83 at 100% drift, coupled with high CPU Time (around 220 seconds). The GRU and LSTM models showed more stable performance in terms of Accuracy but experienced moderate decreases as the drift magnitude increased. The CPU Time for Uncertainty-based Drift Detection was significantly higher compared to the Hybrid Approach, especially for BNN.

- **Gradual Drift:** In the case of Gradual Drift, the Hybrid Approach showed that BiLSTM and BNN models performed consistently well, although CPU Time increased significantly. BiLSTM's Accuracy decreased slightly from 0.86 at a 20% flip to 0.78 at a 100% flip, while BNN followed a similar trend, with Accuracy ranging from 0.81 to 0.77. The CPU Time for these models also escalated as the drift magnitude increased, with BNN requiring the longest processing time. GRU and LSTM models, though less efficient in terms of Accuracy (around 0.78 for higher drift magnitudes), showed more stable performance but still faced increased CPU Time as the drift severity increased. For Gradual Drift, BiLSTM maintained high Accuracy (0.87) at 20% drift and dropped to 0.77 for 100% flip, with a similar trend observed in LSTM. The BNN model struggled significantly, with Accuracy starting at 0.9 for the 20% flip but dropping to 0.77 for the 100% flip, accompanied by high CPU Time. GRU displayed more stability, though its Accuracy was lower compared to BiLSTM and LSTM, starting at 0.86 for 20% flip and declining to 0.78 for 100% flip. The increased CPU Time for BNN indicates that this model struggles under higher drift magnitudes, making it less suitable for real-time applications in such scenarios.
- **Recurring Drift:** For Recurring Drift, the Hybrid Approach showed that BiLSTM continued to outperform the other models, with Accuracy starting at 0.86 for lower drift levels but dropping to 0.72 as the drift magnitude reached 100%. The GRU and LSTM models exhibited the most significant degradation in performance, with Accuracy dropping to around 0.74 to 0.76. Despite the drop in Accuracy, these models maintained similar CPU Time across the different drift levels. The results suggest that while the Hybrid Approach is

TABLE 8. Brier score analysis and calibration of uncertainty in deep learners.

Strategy	Models	Brier Score
Adaptive Buffer Strategy with Uncertainty based Drift Detection	BNN	0.1744
	BiLSTM	0.1743
	LSTM	0.1744
	GRU	0.1757

effective for Recurring Drift, the performance of the models declines as the drift magnitude becomes severe, although BiLSTM remains the most stable model. For Recurring Drift, BiLSTM continued to perform consistently well across different drift magnitudes, with Accuracy starting at 0.86 and dropping to 0.73 for the 100% flip. Similarly, LSTM performed well at 0.81 for 20% drift and slightly decreased to 0.73 at 100% drift. The BNN model exhibited significant deterioration in performance as the drift magnitude increased, with Accuracy dropping from 0.89 at 20% flip to 0.72 at 100% flip, along with increased CPU Time. GRU performed similarly to LSTM, with a moderate drop in Accuracy and consistent CPU Time. Overall, Uncertainty-based Drift Detection performed well but at the cost of higher computational time, especially for BNN, which was less efficient than the Hybrid Approach.

The Uncertainty-based Drift Detection method showed robust performance across all drift types, but the CPU Time was much higher, particularly for BNN. While BiLSTM was the most efficient in terms of Accuracy, models like BNN faced substantial performance degradation with increasing drift magnitude and high CPU Time. In comparison, the Hybrid Approach was more efficient with balanced performance in terms of both Accuracy and CPU Time, especially for Abrupt Drift.

G. CALIBRATION STUDY ON UNCERTAINTY BASED DRIFT DETECTION

Calibration plays a critical role in ensuring that deep learning models produce reliable predicted probabilities that align well with actual outcomes, especially when dealing with uncertainty in concept drift detection. In this study, deep learning models such as BiLSTM, BNN, GRU, and LSTM are utilized for uncertainty estimation, and their predicted probabilities are calibrated to improve decision-making under drift conditions.

The calibration curve, also known as the reliability diagram, provides a visual tool to assess how well the predicted probabilities of a model match the observed outcomes. For perfectly calibrated models, the predicted probabilities should match the observed proportion of positive samples. For example, for a predicted probability of 0.8, the actual proportion of positive samples in that bin should also be 0.8.

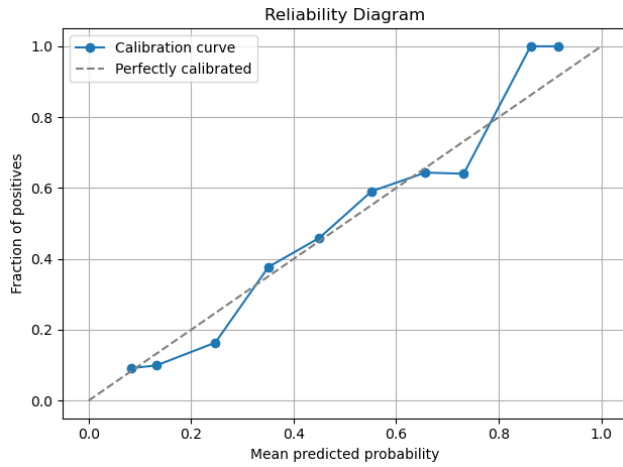


FIGURE 18. Bi-LSTM calibration curve.

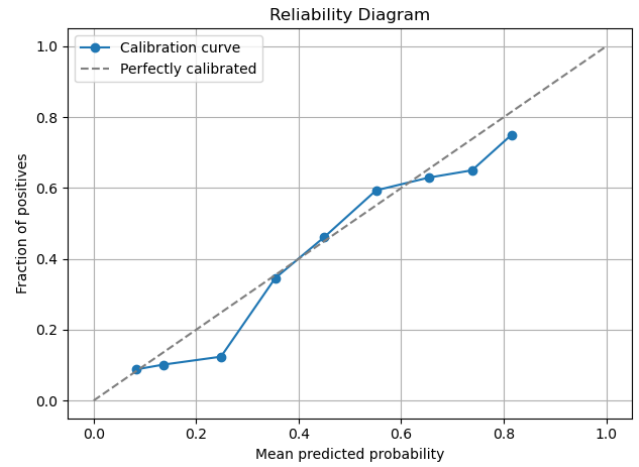


FIGURE 21. GRU calibration curve.

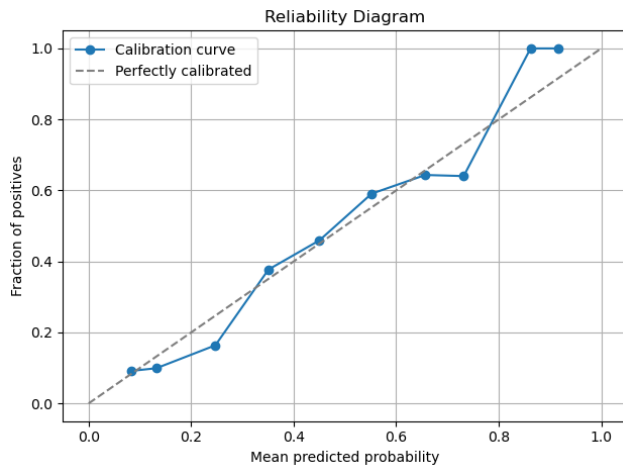


FIGURE 19. BNN calibration curve.

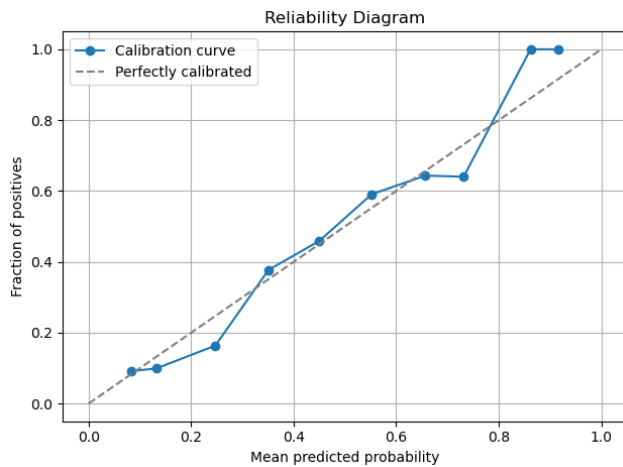


FIGURE 20. LSTM calibration curve.

The calibration curve is computed by grouping the predicted probabilities into bins and comparing the observed

TABLE 9. Retention and forgetting scores for BiLSTM across different drift detection methods.

Drift Detection Method	Buffer Strategy	ARA	FM
Hybrid Drift Detection	No buffer	0.7276	0.0271
	FIFO buffer	0.7318	0.0350
	Adaptive buffer	0.7259	0.0244
Uncertainty-based Drift Detection	No buffer	0.718	0.03
	FIFO buffer	0.718	0.03
	Adaptive buffer	0.71	0.03
KS-test-based Drift Detection	No buffer	0.718	0.03
	FIFO buffer	0.718	0.03
	Adaptive buffer	0.718	0.03

frequency of positive outcomes in each bin with the predicted probability. The formula used to calculate the observed fraction of positives for a given bin is as shown in equation 22 as,

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}(y_i = 1) \quad (22)$$

where: - n is the number of samples in each bin, - $\mathbb{I}(y_i = 1)$ is an indicator function that returns 1 if the true outcome y_i is positive (i.e., 1) and 0 otherwise.

The closer the calibration curve is to the diagonal line, the more accurate and reliable the model's predicted probabilities are.

1) BRIER SCORE

The Brier score is a commonly used metric to quantify how well-calibrated predicted probabilities are. It computes the mean squared difference between the predicted probabilities and the actual outcomes. The Brier score is calculated as shown in equation 23

$$\text{Brier Score} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \quad (23)$$

where: - $\hat{y}_i$ is the predicted probability for the i -th sample, - y_i is the true outcome (0 or 1) for the i -th sample, - N is the total number of samples.

TABLE 10. Performance analysis of deep learners with hybrid drift detection using KS-Test and accuracy dip.

Hybrid Drift Detection using KS-Test and Accuracy dip								
Type of Drift	Buffer Strategy	Models	# Drift Points	Accuracy	Precision	Recall	F1-Score	AUC
Abrupt	Adaptive	BiLSTM	4	0.73	0.71	0.72	0.72	0.78
		BNN	2	0.73	0.71	0.73	0.73	0.78
		GRU	2	0.73	0.71	0.71	0.71	0.78
		LSTM	2	0.73	0.71	0.71	0.71	0.78
	FIFO	BiLSTM	3	0.73	0.71	0.72	0.71	0.78
		BNN	2	0.73	0.71	0.72	0.72	0.78
		GRU	2	0.74	0.72	0.72	0.72	0.78
		LSTM	2	0.73	0.71	0.72	0.72	0.78
	No Buffer	BiLSTM	3	0.73	0.71	0.71	0.71	0.78
		BNN	2	0.73	0.71	0.72	0.71	0.78
		GRU	1	0.73	0.71	0.71	0.71	0.78
Gradual	Adaptive	LSTM	2	0.73	0.71	0.72	0.71	0.78
		BiLSTM	0	0.77	0.75	0.76	0.75	0.84
		BNN	1	0.77	0.75	0.77	0.76	0.84
		GRU	1	0.77	0.74	0.75	0.74	0.84
	FIFO	LSTM	1	0.77	0.75	0.76	0.75	0.84
		BiLSTM	1	0.77	0.75	0.76	0.76	0.84
		BNN	1	0.77	0.75	0.77	0.76	0.84
		GRU	1	0.77	0.75	0.75	0.75	0.84
	No Buffer	LSTM	1	0.77	0.75	0.77	0.76	0.84
		BiLSTM	1	0.77	0.75	0.76	0.76	0.84
		BNN	1	0.77	0.75	0.77	0.76	0.84
		GRU	1	0.76	0.74	0.74	0.74	0.84
Recurring	Adaptive	LSTM	1	0.77	0.75	0.77	0.76	0.84
		BiLSTM	2	0.8	0.8	0.8	0.8	0.85
		BNN	1	0.8	0.8	0.8	0.8	0.85
		GRU	1	0.8	0.8	0.8	0.8	0.85
	FIFO	LSTM	1	0.8	0.8	0.8	0.8	0.85
		BiLSTM	2	0.8	0.8	0.8	0.8	0.85
		BNN	1	0.8	0.8	0.8	0.8	0.85
		GRU	1	0.8	0.8	0.8	0.8	0.85
	No Buffer	LSTM	1	0.8	0.8	0.8	0.8	0.85
		BiLSTM	2	0.8	0.8	0.8	0.8	0.85
		BNN	1	0.8	0.8	0.8	0.8	0.85
		GRU	1	0.8	0.8	0.8	0.8	0.85

TABLE 11. Performance analysis of Deep learners with external drift detection using KS-Test only.

Type of Drift		KS-Test based Drift Detection							
Buffer Strategy	Models	# Drift Points	Accuracy	Precision	Recall	F1-Score	AUC	RunTime(sec)	
Abrupt	Adaptive	BiLSTM	18	0.73	0.71	0.72	0.72	0.78	18.45
		BNN	15	0.73	0.71	0.71	0.71	0.78	27.67
		GRU	18	0.74	0.72	0.72	0.72	0.78	23.25
	FIFO	LSTM	13	0.73	0.71	0.72	0.71	0.78	19.96
		BiLSTM	12	0.72	0.7	0.7	0.7	0.78	35.28
		BNN	15	0.73	0.72	0.73	0.72	0.79	30.015
	No Buffer	GRU	12	0.73	0.72	0.73	0.72	0.78	26.21
		LSTM	15	0.73	0.72	0.73	0.72	0.79	30.06
		BiLSTM	24	0.74	0.72	0.73	0.72	0.78	22
	No Buffer	BNN	16	0.74	0.72	0.72	0.72	0.78	16.48
		GRU	21	0.73	0.71	0.72	0.71	0.78	14.43
		LSTM	16	0.74	0.72	0.72	0.72	0.78	15.62
Gradual	Adaptive	BiLSTM	15	0.78	0.76	0.78	0.76	0.84	14.89
		BNN	15	0.78	0.76	0.77	0.76	0.85	13.81
		GRU	18	0.76	0.74	0.74	0.74	0.84	11.89
	FIFO	LSTM	16	0.78	0.76	0.77	0.76	0.84	20.39
		BiLSTM	15	0.78	0.76	0.78	0.77	0.85	49.1
		BNN	15	0.78	0.76	0.78	0.77	0.85	28.14
	No Buffer	GRU	13	0.78	0.77	0.79	0.77	0.85	22.85
		LSTM	15	0.78	0.76	0.78	0.77	0.85	39.98
		BiLSTM	13	0.77	0.75	0.76	0.75	0.85	20.2
	No Buffer	BNN	14	0.78	0.76	0.78	0.77	0.84	14.39
		GRU	15	0.77	0.75	0.76	0.76	0.84	14.54
		LSTM	14	0.78	0.76	0.78	0.77	0.84	14.54
Recurring	Adaptive	BiLSTM	17	0.8	0.8	0.8	0.8	0.85	33.45
		BNN	18	0.8	0.8	0.8	0.8	0.85	30.06
		GRU	14	0.8	0.8	0.8	0.8	0.85	20.51
	FIFO	LSTM	16	0.8	0.8	0.8	0.8	0.85	22.29
		BiLSTM	14	0.8	0.8	0.8	0.8	0.86	84.7
		BNN	18	0.81	0.81	0.81	0.81	0.85	55.97
	No Buffer	GRU	13	0.8	0.8	0.81	0.8	0.85	47.06
		LSTM	18	0.81	0.81	0.81	0.81	0.85	56.12
		BiLSTM	22	0.81	0.81	0.81	0.81	0.85	55.59
	No Buffer	BNN	14	0.8	0.8	0.8	0.8	0.85	35.4
		GRU	17	0.8	0.8	0.8	0.8	0.85	29.48
		LSTM	14	0.8	0.8	0.8	0.8	0.85	31.35

TABLE 12. Performance analysis of Deep learners with uncertainty based drift detection.

Uncertainty based Drift Detection									
Type of Drift	Buffer Strategy	Models	# Drift Points	Accuracy	Precision	Recall	F1-Score	AUC	RunTime(sec)
Abrupt	Adaptive	BiLSTM	14	0.73	0.71	0.71	0.71	0.78	28.5
		BNN	9	0.7	0.68	0.66	0.67	0.76	233.01
		GRU	19	0.72	0.7	0.69	0.7	0.78	18
	FIFO	LSTM	21	0.73	0.71	0.71	0.71	0.78	24.48
		BiLSTM	17	0.73	0.71	0.72	0.71	0.79	50.42
		BNN	9	0.71	0.69	0.68	0.69	0.77	363.45
	No Buffer	GRU	16	0.73	0.71	0.72	0.72	0.79	302.76
		LSTM	21	0.74	0.72	0.74	0.73	0.78	37.79
		BiLSTM	18	0.73	0.72	0.72	0.72	0.79	36.26
Gradual	Adaptive	BNN	16	0.73	0.71	0.72	0.72	0.78	11.95
		GRU	19	0.73	0.71	0.71	0.71	0.78	17.95
		LSTM	21	0.73	0.71	0.72	0.71	0.78	26.92
	FIFO	BiLSTM	17	0.78	0.76	0.77	0.76	0.84	40.85
		BNN	9	0.69	0.66	0.63	0.64	0.76	233.0156
		GRU	15	0.77	0.75	0.76	0.76	0.85	23.76
	No Buffer	LSTM	20	0.77	0.76	0.77	0.76	0.84	25.84
		BiLSTM	17	0.78	0.76	0.77	0.76	0.85	60.73
		BNN	10	0.78	0.76	0.78	0.77	0.85	307.21
Recurring	Adaptive	GRU	11	0.78	0.76	0.77	0.76	0.86	358.51
		LSTM	20	0.78	0.76	0.77	0.76	0.85	49.89
		BiLSTM	12	0.78	0.76	0.77	0.76	0.84	33.93
	No Buffer	BNN	14	0.78	0.77	0.79	0.77	0.85	19.2
		GRU	15	0.77	0.75	0.76	0.75	0.84	22.76
		LSTM	16	0.78	0.76	0.78	0.77	0.84	25.75
	Adaptive	BiLSTM	19	0.8	0.8	0.8	0.8	0.86	40.68
		BNN	9	0.69	0.66	0.63	0.64	0.76	233.01
		GRU	15	0.8	0.8	0.8	0.8	0.85	24.12
FIFO	Adaptive	LSTM	21	0.8	0.8	0.8	0.8	0.85	27.06
		BiLSTM	19	0.8	0.81	0.8	0.8	0.86	66.09
		BNN	6	0.83	0.83	0.83	0.83	0.86	246.15
	No Buffer	GRU	14	0.82	0.82	0.82	0.82	0.87	294.67
		LSTM	21	0.81	0.81	0.81	0.81	0.86	46.95
		BiLSTM	17	0.81	0.81	0.81	0.81	0.85	36.18
	No Buffer	BNN	12	0.81	0.81	0.81	0.81	0.85	17.56
		GRU	15	0.8	0.8	0.8	0.8	0.85	20.92
		LSTM	16	0.8	0.8	0.8	0.8	0.85	27.04

A lower Brier score indicates a more reliable model that accurately predicts the likelihood of an event. Models that exhibit lower Brier scores across different drift types demonstrate better calibration, which is crucial when detecting and responding to concept drift. The Brier score measures the accuracy of probabilistic predictions, quantifying the mean squared difference between predicted probabilities and actual outcomes and table 8 summarizes the findings. The BiLSTM model outperforms the other models by having the lowest Brier score of 0.1743, which indicates that it produces the most reliable predicted probabilities among the four models. BNN and LSTM have almost identical Brier scores (0.1744), showing that their probabilistic predictions are similarly calibrated. The GRU model has the highest Brier score of 0.1757, though still within a reasonable range, suggesting slightly less reliable calibration compared to the other models.

The calibration curve for BNN as shown in the figure 19 that the model tends to be slightly overconfident at both low and high probabilities. It deviates from the perfectly calibrated line, especially at extreme ends. The Brier score for BNN was the highest among the models, indicating relatively poor calibration compared to others. The curve for BiLSTM is closer to the perfectly calibrated line, suggesting better calibration compared to the other models. BiLSTM's Brier score as shown in figure 18 was one of the lowest, indicating a better calibration and more reliable uncertainty predictions. The calibration curve for GRU shown in 21 also shows a reasonable match with the perfectly calibrated line, though it is slightly more spread out than BiLSTM. GRU's Brier score was slightly higher than BiLSTM, indicating that it performs slightly worse in terms of calibration. LSTM's calibration curve as shown in figure 20 shows a near-perfect alignment with the ideal line, indicating the best calibration among the four models. LSTM had the lowest Brier score, making it the best in terms of calibration.

H. CATASTROPHIC FORGETTING AND RETENTION

Catastrophic forgetting refers to the phenomenon where a model, after being trained on new data, loses its ability to perform well on previously learned tasks. In the context of concept drift, this is particularly relevant because the model may forget its knowledge of earlier data distributions as it is exposed to new data distributions. This can result in a significant drop in performance on earlier tasks. As a case study we have taken BiLSTM model with various drift detection methods (Hybrid, Uncertainty based and Statistical test based) against the three buffers (Adaptive, FIFO, and No-buffer) and studied how the proposed methods avoids the catastrophic forgetting during its runtime. Two measures are used in this study, they are,

- 1) **Catastrophic Forgetting (FM)** is computed as the difference between the maximum accuracy on a task and the final accuracy after new tasks have been

learned. FM can be calculated as equation 24,

$$FM = \frac{1}{N} \sum_{i=1}^N (A_i^{\max} - A_i^{\text{final}}) \quad (24)$$

where: - N is the number of tasks (or batches) in the archive. - $A_i^{\max}$ is the highest accuracy achieved on task i . - A_i^{final} is the accuracy on task i after the final training. A low FM value indicates minimal forgetting, meaning the model retains knowledge from older tasks. A high FM value indicates catastrophic forgetting, where the model's performance deteriorates significantly on older tasks as it learns new ones.

- 2) **Retention Accuracy (ARA)**: Retention Accuracy (ARA) is a metric used to quantify how well the model retains performance on earlier tasks as it is exposed to new data. It is the average accuracy on the older tasks after the model has been trained on newer tasks. Let A_{t-k} be the accuracy on task $t - k$ after training on task t . The Retention Accuracy (ARA) for the model after training on task t is defined as the average accuracy on all previous tasks $t - k$, where k is the number of batches or tasks before t as equation 25,

$$ARA_{t-k} = \frac{1}{K} \sum_{i=1}^K A_{t-k} \quad (25)$$

where: - A_{t-k} is the accuracy on task $t - k$ after training up to task t . - K is the total number of previous tasks (or batches) to evaluate.

ARA measures how well the model continues to perform on previous tasks after learning from new data. A high ARA indicates that the model has retained knowledge of previous tasks, and thus, it is less prone to catastrophic forgetting.

The table 9 compares the performance of BiLSTM (Bidirectional Long Short-Term Memory) in terms of retention (ARA) and forgetting (F) across different drift detection methods and buffer strategies. The Hybrid Drift Detection method with the FIFO buffer results in the highest retention (ARA = 0.7318) but also the highest forgetting rate (0.0350). This suggests that while the FIFO buffer slightly improves retention, it also leads to more forgetting compared to other methods. The Uncertainty-based Drift Detection method and KS-test-based Drift Detection yield similar results, with no significant change in retention and forgetting across buffer strategies. However, adaptive buffering helps in reducing the forgetting rate compared to FIFO and no buffer scenarios. Adaptive buffering generally reduces forgetting rates across all methods, but it also comes with slight decreases in retention, particularly in Uncertainty-based drift detection. BiLSTM performs reasonably well with the Hybrid Drift Detection method, especially with the FIFO buffer, but the adaptive buffer seems to be the best in terms of minimizing forgetting, though it slightly affects retention.

TABLE 13. Ablations for hybrid drift detection using deep learners with adaptive buffer strategy with different drift types(Abrupt, Gradual and Recurring).

Hybrid Drift Detection with adaptive buffering									
Type of Drift	Ablation	Models	# Drift Points	Accuracy	Precision	Recall	F1-Score	AUC	RunTime(sec)
Abrupt	B1	BiLSTM	4	0.73	0.71	0.72	0.72	0.78	24.29
		BNN	2	0.73	0.71	0.73	0.73	0.78	23.79
		GRU	2	0.73	0.71	0.71	0.71	0.78	9.95
		LSTM	2	0.73	0.71	0.71	0.71	0.78	15.09
	B2	BiLSTM	1	0.82	0.82	0.82	0.82	0.87	145.01
		BNN	2	0.82	0.82	0.82	0.82	0.87	51.82
		GRU	2	0.82	0.82	0.82	0.82	0.86	42.93
		LSTM	2	0.82	0.82	0.82	0.82	0.87	96.17
	B3	BiLSTM	2	0.81	0.81	0.81	0.81	0.86	15.7
		BNN	1	0.8	0.8	0.8	0.8	0.85	8.35
		GRU	2	0.81	0.81	0.81	0.81	0.86	7.73
		LSTM	1	0.8	0.8	0.8	0.8	0.85	8.64
	B4	BiLSTM	13	0.82	0.82	0.82	0.82	0.86	31.23
		BNN	13	0.81	0.81	0.81	0.81	0.86	33.37
		GRU	14	0.81	0.81	0.81	0.81	0.86	32.21
		LSTM	13	0.81	0.81	0.81	0.81	0.86	33.67
Gradual	B1	BiLSTM	0	0.77	0.75	0.76	0.75	0.84	22.45
		BNN	1	0.77	0.75	0.77	0.76	0.84	24.5
		GRU	1	0.77	0.74	0.75	0.74	0.84	15.28
		LSTM	1	0.77	0.75	0.76	0.75	0.84	16.12
	B2	BiLSTM	2	0.79	0.77	0.79	0.78	0.85	77.42
		BNN	0	0.79	0.77	0.79	0.78	0.85	59.53
		GRU	2	0.79	0.77	0.79	0.77	0.85	52.87
		LSTM	0	0.79	0.77	0.79	0.78	0.85	59.37
	B3	BiLSTM	1	0.78	0.76	0.77	0.76	0.85	17.85
		BNN	1	0.78	0.76	0.77	0.76	0.85	10.15
		GRU	0	0.77	0.75	0.76	0.75	0.85	9.01
		LSTM	1	0.78	0.76	0.77	0.76	0.85	9.62
	B4	BiLSTM	13	0.79	0.76	0.78	0.77	0.85	23.87
		BNN	15	0.79	0.77	0.79	0.77	0.85	17.48
		GRU	15	0.78	0.76	0.77	0.76	0.85	16.39
		LSTM	15	0.79	0.77	0.79	0.77	0.85	17.76
Recurring	B1	BiLSTM	2	0.8	0.8	0.8	0.8	0.85	25.65
		BNN	1	0.8	0.8	0.8	0.8	0.85	24.1
		GRU	1	0.8	0.8	0.8	0.8	0.85	16.4
		LSTM	1	0.8	0.8	0.8	0.8	0.85	15.51
	B2	BiLSTM	2	0.74	0.74	0.74	0.74	0.78	79.67
		BNN	1	0.74	0.74	0.74	0.74	0.78	74.26
		GRU	3	0.74	0.74	0.74	0.74	0.78	74.48
		LSTM	1	0.74	0.74	0.74	0.74	0.78	56.15
	B3	BiLSTM	2	0.73	0.73	0.73	0.73	0.77	28.15
		BNN	2	0.73	0.73	0.73	0.73	0.77	16.9
		GRU	2	0.73	0.73	0.73	0.73	0.77	15.21
		LSTM	2	0.73	0.73	0.73	0.73	0.77	17.1
	B4	BiLSTM	17	0.73	0.74	0.73	0.73	0.77	44.2
		BNN	16	0.73	0.74	0.73	0.73	0.77	22.6
		GRU	17	0.73	0.74	0.73	0.73	0.77	24.29
		LSTM	16	0.73	0.74	0.73	0.73	0.77	26.71

XI. CLINICAL INSIGHTS AND CASE STUDIES

The integration of drift-aware incremental learning techniques with lifestyle disease prediction offers meaningful insights for real-world clinical decision support. In this section, we discuss practical implications derived from the experimental findings and potential case-based applications that align with public health surveillance and individual patient care.

A. PERSONALIZED MONITORING AND EARLY DETECTION

The findings through presented experimentation suggest that adaptive buffering mechanisms combined with uncertainty-aware detection can track subtle patient behavior changes over time. For example, a shift in dietary intake or

physical activity pattern—common precursors to metabolic dysregulation can be captured as concept drift. Such timely detection allows clinicians to intervene proactively, potentially preventing disease progression.

B. EHR-INTEGRATED LEARNING SYSTEMS

The NHANES-based models simulated a streaming environment analogous to real-world Electronic Health Records (EHRs). In a clinical deployment, the models can continuously learn from patient-specific data and issue alerts when a patient's risk profile deviates from historical patterns. These alerts, supported by high model uncertainty, serve as justifiable triggers for additional diagnostics or lifestyle counseling.

TABLE 14. Ablations for uncertainty based drift detection using deep learners with adaptive buffer strategy with different drift types(Abrupt, Gradual and Recurring).

Uncertainty based Drift Detection									
Type of Drift	Ablation	Models	# Drift Points	Accuracy	Precision	Recall	F1-Score	AUC	RunTime(sec)
Abrupt	U1	BiLSTM	14	0.73	0.71	0.71	0.71	0.78	28.5
		BNN	9	0.7	0.68	0.66	0.67	0.76	233.01
		GRU	19	0.72	0.7	0.69	0.7	0.78	18
		LSTM	21	0.73	0.71	0.71	0.71	0.78	24.48
	U2	BiLSTM	0	0.82	0.82	0.82	0.82	0.86	86.93
		BNN	0	0.82	0.82	0.82	0.82	0.86	57.4
		GRU	0	0.81	0.82	0.82	0.81	0.86	49.95
		LSTM	0	0.82	0.82	0.82	0.82	0.86	45.32
	U3	BiLSTM	0	0.81	0.81	0.81	0.81	0.86	24.09
		BNN	0	0.8	0.8	0.81	0.8	0.86	15.29
		GRU	0	0.81	0.81	0.81	0.81	0.86	13.6
		LSTM	0	0.81	0.81	0.81	0.81	0.86	11.59
	U4	BiLSTM	32	0.8	0.8	0.81	0.8	0.86	39.57
		BNN	36	0.81	0.81	0.81	0.81	0.85	25.67
		GRU	26	0.81	0.81	0.81	0.81	0.86	23.98
		LSTM	23	0.81	0.82	0.82	0.81	0.86	18.21
	U5	BiLSTM	32	0.8	0.8	0.81	0.8	0.85	63.84
		BNN	36	0.8	0.8	0.8	0.8	0.85	42.53
		GRU	26	0.81	0.81	0.81	0.81	0.86	34.48
		LSTM	20	0.81	0.81	0.81	0.81	0.85	26.93
Gradual	U1	BiLSTM	17	0.78	0.76	0.77	0.76	0.84	40.85
		BNN	9	0.69	0.66	0.63	0.64	0.76	233.0156
		GRU	15	0.77	0.75	0.76	0.76	0.85	23.76
		LSTM	20	0.77	0.76	0.77	0.76	0.84	25.84
	U2	BiLSTM	0	0.78	0.76	0.77	0.76	0.85	108
		BNN	0	0.79	0.77	0.78	0.77	0.86	0.86
		GRU	0	0.78	0.76	0.76	0.76	0.85	55.75
		LSTM	0	0.79	0.77	0.79	0.77	0.85	64.78
	U3	BiLSTM	0	0.78	0.75	0.77	0.76	0.85	11.84
		BNN	0	0.79	0.77	0.79	0.77	0.85	7.64
		GRU	0	0.77	0.75	0.76	0.75	0.85	8.26
		LSTM	0	0.78	0.76	0.77	0.76	0.85	8.32
	U4	BiLSTM	27	0.78	0.76	0.77	0.76	0.85	24.98
		BNN	25	0.79	0.77	0.79	0.77	0.85	12.37
		GRU	21	0.77	0.75	0.76	0.75	0.85	14.82
		LSTM	32	0.78	0.76	0.77	0.76	0.85	18.12
	U5	BiLSTM	27	0.75	0.73	0.74	0.73	0.83	35.46
		BNN	25	0.75	0.72	0.72	0.72	0.83	18.65
		GRU	21	0.75	0.73	0.74	0.73	0.83	18.18
		LSTM	32	0.75	0.73	0.74	0.73	0.83	21.15
Recurring	U1	BiLSTM	19	0.8	0.8	0.8	0.8	0.86	40.68
		BNN	9	0.69	0.66	0.63	0.64	0.76	233.01
		GRU	15	0.8	0.8	0.8	0.8	0.85	24.12
		LSTM	21	0.8	0.8	0.8	0.8	0.85	27.06
	U2	BiLSTM	0	0.73	0.73	0.73	0.73	0.77	63.29
		BNN	0	0.72	0.73	0.73	0.72	0.76	35.07
		GRU	0	0.73	0.73	0.73	0.73	0.77	35.34
		LSTM	0	0.73	0.73	0.73	0.73	0.77	36.25
	U3	BiLSTM	0	0.73	0.73	0.73	0.73	0.77	27.81
		BNN	0	0.73	0.73	0.73	0.73	0.77	13.6
		GRU	0	0.73	0.73	0.73	0.73	0.77	15.42
		LSTM	0	0.73	0.73	0.73	0.73	0.77	15.73
	U4	BiLSTM	34	0.73	0.73	0.73	0.73	0.77	31.96
		BNN	28	0.73	0.73	0.73	0.73	0.77	16.89
		GRU	28	0.73	0.73	0.73	0.73	0.77	17.28
		LSTM	35	0.73	0.73	0.73	0.73	0.77	17.71
	U5	BiLSTM	34	0.71	0.71	0.71	0.71	0.75	43.2
		BNN	28	0.72	0.72	0.72	0.72	0.75	23.5
		GRU	28	0.73	0.73	0.73	0.73	0.77	24.01
		LSTM	35	0.72	0.72	0.72	0.72	0.76	26.68

C. PUBLIC HEALTH SURVEILLANCE

On a population scale, the model’s ability to detect population-level drifts such as seasonal trends in hypertension or glucose levels can support public health strategies.

For instance, sudden increases in drift points across multiple models may indicate community-wide behavioral changes or environmental exposures, warranting targeted health campaigns.

TABLE 15. Study on drift magnitude and its impact on performance for various drift types(Abrupt, Gradual and recurring) with adaptive buffer strategy using various deep learners.

Drift Magnitude			20% Flip			50% Flip			80% Flip			100% Flip		
Drift Detection	Drift type	Models	# Drift Points	Accuracy	CPU Time	# Drift Points	Accuracy	CPU Time	# Drift Points	Accuracy	CPU Time	# Drift Points	Accuracy	CPU Time
Hybrid Approach	Abrupt	BiLSTM	1	0.94	5.5	3	0.85	16.17	3	0.83	15.23	2	0.81	12.56
		BNN	0	0.93	9.2	0	0.85	23.17	1	0.82	22.65	1	0.81	24.57
		GRU	1	0.95	8.07	1	0.85	14.87	3	0.83	16.97	3	0.81	15.57
		LSTM	1	0.94	9.4	1	0.84	16.04	3	0.82	16.95	2	0.81	16.45
	Gradual	BiLSTM	1	0.86	26.46	3	0.83	25.57	1	0.81	23.23	0	0.78	22.45
		BNN	1	0.86	27.6	1	0.82	23.45	1	0.8	23.46	1	0.78	24.5
		GRU	1	0.86	18.79	1	0.82	14.95	1	0.81	15.71	1	0.78	15.28
		LSTM	0	0.86	22.18	3	0.82	11.9	2	0.81	11.28	1	0.78	11.28
	Recurring	BiLSTM	2	0.86	35.56	4	0.81	32.26	3	0.76	26.71	2	0.73	25.65
		BNN	2	0.86	28.28	3	0.81	24.28	3	0.76	24.04	3	0.73	24.1
		GRU	1	0.86	19.76	2	0.81	15.81	3	0.76	16.06	3	0.73	16.4
		LSTM	2	0.86	16.4	2	0.81	16.51	3	0.76	17.7	1	0.73	15.51
Uncertainty based Drift Detection	Abrupt	BiLSTM	15	0.87	30.45	18	0.85	28.29	17	0.82	25.84	21	0.8	28.2
		BNN	9	0.9	220.6	6	0.88	215.89	8	0.85	231.2	11	0.83	220.29
		GRU	14	0.87	19.35	13	0.85	15.5	14	0.83	16.67	14	0.8	15.78
		LSTM	20	0.88	22.21	21	0.85	18.92	21	0.83	18.23	21	0.81	18.92
	Gradual	BiLSTM	16	0.87	30.7	18	0.82	26.15	18	0.81	26.1	17	0.77	24.34
		BNN	10	0.9	230.48	7	0.83	435.85	1	0.79	403.97	3	0.77	410.1
		GRU	12	0.86	30.5	13	0.81	28.79	14	0.8	29.43	14	0.76	30.09
		LSTM	18	0.86	35.75	19	0.83	35	18	0.81	33.2	21	0.78	33.5
	Recurring	BiLSTM	17	0.86	48.48	19	0.81	47.17	19	0.76	48.51	21	0.73	48.65
		BNN	11	0.89	439.29	13	0.81	415.9	11	0.75	416.92	8	0.72	414.07
		GRU	11	0.86	31.71	13	0.8	27.45	17	0.75	30.09	19	0.73	29.81
		LSTM	19	0.86	39.42	22	0.81	35.54	21	0.76	34.89	20	0.73	35.46
KS-Test based Drift Detection	Abrupt	BiLSTM	17	0.88	22.96	17	0.85	17.07	19	0.83	18.1	18	0.81	18.45
		BNN	20	0.87	27.25	18	0.85	22.01	19	0.82	22.21	16	0.81	21.45
		GRU	18	0.88	19.03	17	0.85	14.75	18	0.82	15.06	12	0.81	13.45
		LSTM	19	0.87	21.03	19	0.85	16.67	22	0.83	16.81	23	0.81	18.6
	Gradual	BiLSTM	12	0.86	19.12	14	0.83	16.01	11	0.8	14.6	13	0.78	14.89
		BNN	17	0.86	19.95	17	0.83	15.07	14	0.8	14.84	13	0.77	13.81
		GRU	12	0.86	14.96	18	0.83	11.95	14	0.8	10.79	19	0.78	11.89
		LSTM	18	0.86	19.43	17	0.83	16.07	17	0.81	15.96	14	0.77	14.03
	Recurring	BiLSTM	20	0.86	46.39	19	0.81	35.1	15	0.76	31.98	17	0.72	33.45
		BNN	21	0.86	35.39	17	0.81	30	18	0.75	30.9	17	0.73	30.06
		GRU	13	0.86	22.98	17	0.81	21.92	16	0.74	21.1	14	0.72	20.51
		LSTM	21	0.86	22.39	19	0.81	19.89	15	0.76	15	16	0.73	15.29

D. CASE EXAMPLE: DRIFT IN PHYSICAL ACTIVITY AND DIABETES ONSET

Consider a middle-aged patient whose physical activity sharply declines over several weeks, as captured by wearable devices. The GRU model with adaptive buffering detects a drift in the 'PAQ605' (work-related activity) and 'PAQ620' (recreational activity) features, correlated with rising BMI and fasting glucose levels. The system flags this shift with high uncertainty, prompting the clinician to assess the risk of diabetes onset and recommend behavioral modifications.

E. TRUSTWORTHY AI IN MEDICINE

The use of Bayesian Neural Networks (BNNs) not only improved drift detection but also introduced a quantifiable uncertainty metric, an essential feature in clinical AI systems. This uncertainty-aware modeling enhances trust, allowing clinicians to interpret predictions with confidence thresholds and avoid over-reliance on automated decisions.

F. RECOMMENDATIONS FOR CLINICAL DEPLOYMENT

- Employ adaptive buffer-based learners to balance historical context and real-time updates.
- Use uncertainty scores as a gating mechanism for generating clinical alerts.
- Integrate statistical drift logs into EHR systems for transparent auditing and explainability.
- Perform periodic model validation to ensure alignment with shifting population health dynamics.

In summary, the proposed framework not only supports continuous learning but also aligns well with precision health goals, enabling systems that evolve with the patient and population alike.

XII. CONCLUSION AND FUTURE SCOPE

In this study on Adaptive Buffering Strategies for Incremental Learning Under Concept Drift in Lifestyle Disease Modeling, four deep learning models: Bi-LSTM, LSTM, BNN, and GRU were evaluated for concept drift detection using three distinct buffering strategies (Adaptive buffer, FIFO buffer, and No-buffer) and three types of drift detection mechanisms (KS-test, Hybrid drift detection, and Uncertainty-based drift detection). The analysis revealed that adaptive buffering strategies consistently outperformed FIFO and No-buffer strategies across all models, maintaining higher accuracy, precision, and recall, particularly under Abrupt, Gradual, and Recurring drifts. Hybrid drift detection provided the best overall performance, with Bi-LSTM showing strong retention and minimal forgetting, especially in the face of high drift magnitudes. The drift magnitude study demonstrated that increasing the intensity of drift (from 20% to 100%) significantly impacted model performance, with models employing adaptive buffering and uncertainty-based drift detection proving more resilient in handling large drifts. Notably, BNN struggled with high drift magnitudes, underscoring the importance of combining robust drift detection methods with effective adaptation strategies like retraining. Overall,

the study emphasizes the critical role of adaptive buffering strategies and advanced drift detection techniques in ensuring effective and accurate incremental learning in the presence of concept drift.

FUTURE SCOPE

The following directions are envisioned to extend the utility and scope of this framework:

- **Multi-modal Learning:** Integrate imaging, sensor, and genomic data to enhance feature space and increase drift sensitivity.
- **Explainable Buffering:** Design interpretable buffer strategies to explain sample retention or replacement decisions for clinical auditability.
- **Federated Deployment:** Expand to privacy-aware federated learning settings across hospitals and demographics, ensuring real-time drift tracking without data centralization.
- **Dynamic Policy Triggers:** Implement real-time, uncertainty-aware drift alarms and model re-calibration triggers tailored to disease progression patterns.
- **Clinical Validation:** Collaborate with healthcare providers to deploy and evaluate the framework in live electronic health record (EHR) environments for real-world validation.

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